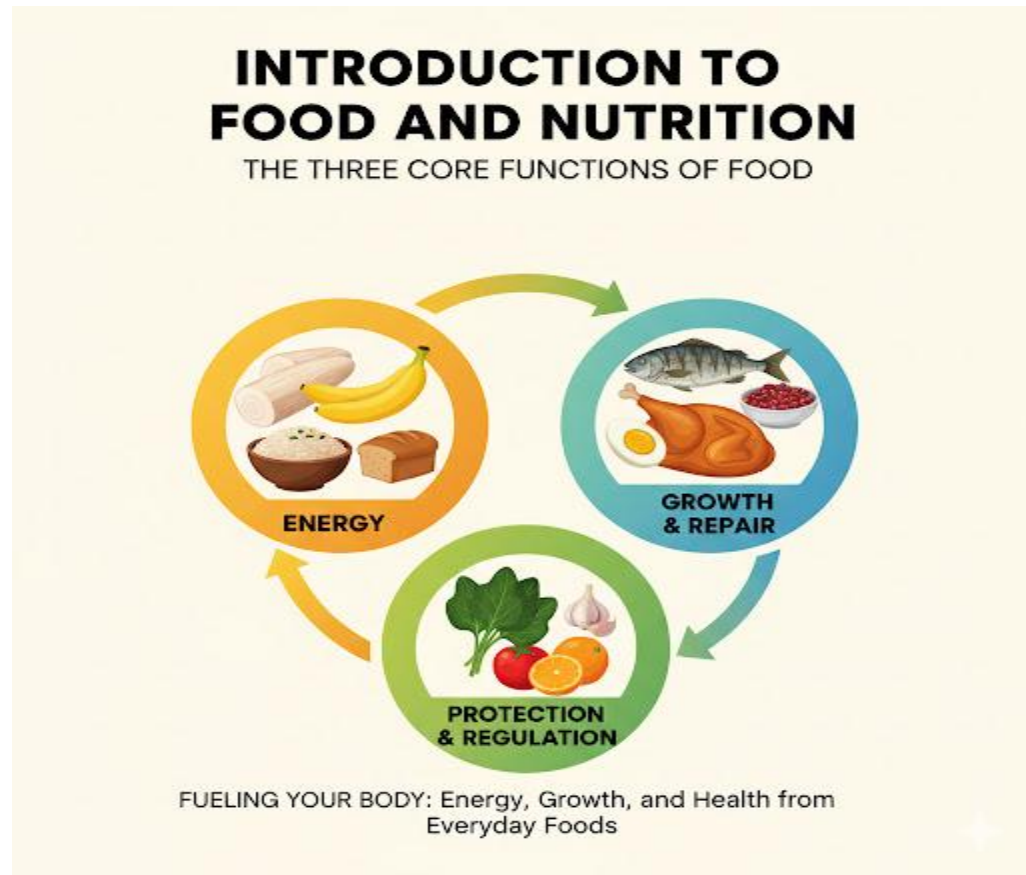


SS1 FOOD AND NUTRITION LESSON NOTES

FIRST TERM:

WEEK 1: INTRODUCTION TO FOOD AND NUTRITION

Topic: Basic Terms, Importance, and Factors Affecting Food Choices



1. Definition of Basic Terms

- **Food:** Any substance, solid or liquid (e.g., yam, water, milk), which when eaten, digested, and absorbed into the body, provides nourishment. It is differentiated from "foodstuffs," which are the raw materials (e.g., raw cassava).
 - **Core Functions:**
 1. **Provides Energy (Fuel):** For all voluntary and involuntary actions (e.g., running, breathing, heartbeat).

2. **Promotes Growth and Repair (Building):** Provides materials to build new cells, replace old ones, and repair damaged tissues.
 3. **Protects and Regulates (Protection):** Maintains the immune system, helps in recovery from illness, and regulates body processes.
- **Nutrition:** The scientific study of food, the nutrients contained within, and the entire process of how the body uses them. This includes the six stages: **Ingestion** (eating), **Digestion** (breaking down), **Absorption** (passing into the blood), **Transport** (moving to cells), **Metabolism** (using in cells), and **Excretion** (removing waste). It also encompasses the study of the social, economic, and cultural factors that influence food choices and health outcomes.
 - **Nutrients:** The specific chemical substances found in food that are essential for life and growth. They are categorized into six classes: Carbohydrates, Proteins, Fats & Oils (Lipids), Vitamins, Minerals, and Water.
 - **Malnutrition:** "Bad" or "poor" nutrition. A comprehensive term for an imbalance (deficiency, excess, or wrong ratio) in a person's intake of energy and/or nutrients.
 - **Undernutrition:** The most common form in developing nations. Occurs when the diet lacks sufficient quantity (energy) or quality (nutrients).
 - **Examples:** Kwashiorkor (protein deficiency), Anaemia (iron deficiency), Marasmus (energy & protein deficiency).
 - **Overnutrition:** A growing problem globally. Occurs from an excessive intake of food, especially energy-dense, low-nutrient foods.
 - **Examples:** Obesity, Type 2 Diabetes, Hypertension, Heart Disease.
 - **Iatrogenic Malnutrition:** A less common form caused unintentionally by medical treatment (e.g., long-term hospital stays on liquid diets, side effects of medication that reduce appetite).
 - **Antinutrient:** A substance naturally found in some foods (especially plants) that can interfere with the absorption or utilization of other nutrients.
 - **Mechanisms:** They can bind to nutrients, making them insoluble, or inhibit enzymes needed for digestion.
 - **Examples:**

- **Oxalates** (in spinach, *ugwu*, cocoa) bind with calcium, forming insoluble calcium oxalate, which the body cannot absorb.
- **Phytates/Phytic Acid** (in whole grains, beans, nuts) reduce the absorption of iron, zinc, and calcium.
- **Trypsin Inhibitors** (in raw soybeans, raw eggs - avidin) block the enzyme *trypsin*, which is essential for digesting protein.
- **Goitrogens** (in raw cabbage, cassava) can interfere with iodine uptake, potentially leading to goitre.
 - **Mitigation:** These are often reduced or destroyed by proper processing like **soaking, fermenting (e.g., *fufu*), and thorough cooking.**
- **Pro-vitamin:** A "precursor" substance that is not a vitamin itself but can be converted by the body into an active vitamin.
 - **Examples:**
 0. **Beta-carotene** (the orange/red pigment in red palm oil, carrots, mangoes, pawpaw, and dark green leaves like *ugwu*) is a pro-vitamin for **Vitamin A**.
 1. The amino acid **Tryptophan** (from protein foods) can be converted by the body into the B-vitamin **Niacin (B3)**.
- **Diet:** The specific collection of foods and drinks a person or group consumes regularly.
 - **Therapeutic Diet:** A special diet prescribed by a dietitian to help treat a medical condition (e.g., a low-sodium diet for high blood pressure).
- **Balanced Diet:** A diet that provides all six classes of nutrients in the correct proportions (based on an individual's age, sex, activity level, and health status) to meet the body's needs for energy, growth, and protection, without leading to deficiency or excess. This is often guided by **Recommended Dietary Allowances (RDAs)**.

2. Importance of Food and Nutrition

1. **For Energy:** Food (primarily carbohydrates and fats) provides the chemical energy (measured in Kilocalories or Kilojoules) required for all metabolic processes, physical activity, and maintaining body temperature.

2. **For Growth and Repair:** Protein is the primary "building block" for new cells, muscles, organs (e.g., during childhood, pregnancy) and for the constant repair of tissues after injury or normal wear-and-tear.
3. **For Protection and Regulation:** Vitamins and minerals act as "spark plugs" (co-enzymes) for chemical reactions. They boost the immune system (Vit. C, Zinc), aid vision (Vit. A), build bones (Calcium, Vit. D), transport oxygen (Iron), and ensure proper nerve function (Sodium, Potassium).
4. **For Disease Prevention:** Good nutrition is the best defense against:
 - **Deficiency Diseases:** (e.g., scurvy, rickets, goitre, kwashiorkor).
 - **Chronic "Lifestyle" Diseases:** (e.g., hypertension, type 2 diabetes, obesity, some cancers).
5. **For Mental Well-being:** A well-nourished body and brain functions better. Good nutrition can improve concentration, reduce irritability, and is linked to better mental health.
6. **For Social & Cultural Expression:** Food is a central part of cultural identity, celebrations (weddings, festivals), and social bonding.

3. Factors Affecting Food and Nutrition

1. **Economic Factors (Available Income):** This is often the primary constraint. Low income limits the *quantity*, *quality*, and *variety* of food, often forcing a reliance on cheap, starchy staples (like *gari*, *akpu*) and limiting access to expensive proteins and fresh fruits/vegetables.
2. **Geographical/Environmental Factors:**
 - **Availability:** You eat what is locally available. Riverine areas have more fish; savanna areas have more grains (millet, sorghum).
 - **Climate & Soil:** Determine which crops can be grown.
 - **Storage/Infrastructure:** Lack of good roads, refrigeration, or silos means food (especially fresh produce) can spoil before reaching markets, leading to scarcity and high prices.
3. **Seasonal Factors:** Seasonal foods (like mangoes, corn, *ube*) are cheaper, fresher, and more nutritious when in season. Out-of-season, they are scarce and expensive.

4. Cultural and Religious Factors:

- **Culture:** Traditions and taboos dictate what is "food" and what is "not food" (e.g., insects), how food is prepared (e.g., *fufu*), and what foods are symbolic (e.g., kola nut).
- **Religion:** Some religions have strict food laws (e.g., Islam forbids pork and alcohol; Hinduism forbids beef; Seventh-Day Adventists often follow a vegetarian diet).

5. Social Factors (Food Habits & Preferences):

- **Family:** You learn your primary food habits from your family.
- **Peer Pressure:** Especially among teenagers, wanting to eat what friends are eating (e.g., fast food, snacks).
- **Prestige/Status:** Some foods are seen as "high status" (e.g., imported rice, packaged foods), while others are seen as "poor people's food."

6. **Nutritional Knowledge/Education:** Knowing what constitutes a balanced diet (e.g., the value of combining beans and rice) allows a person to make better choices, *even* with a limited budget. Lack of knowledge can lead to poor choices (e.g., feeding a child only starchy paps).

7. **Time and Convenience:** In modern urban life, busy work schedules mean less time for cooking. This leads to a reliance on "convenience" foods (e.g., instant noodles) and "fast food," which are often high in salt, sugar, and saturated fat.

8. Psychological Factors:

- **Stress/Mood:** People may overeat ("comfort eating") or undereat when stressed, sad, or bored.
- **Advertising & Media:** Aggressive marketing (especially to children) can create a high demand for processed snacks and sugary drinks.

4. Evaluation (Week 1)

• Objective Questions:

1. Any substance, solid or liquid, that provides nourishment to the body is called: (a) Nutrition (b) Nutrient (c) Food (d) Diet.

2. The scientific study of food and how the body uses it is: (a) Nutrition (b) Dietetics (c) Food Science (d) Malnutrition.
3. A substance in *ugwu* leaves (oxalate) that can block calcium absorption is an: (a) Pro-vitamin (b) Antinutrient (c) Amino acid (d) Enzyme.
4. Beta-carotene, which the body converts to Vitamin A, is an example of a: (a) Pro-vitamin (b) Antinutrient (c) Fat (d) Protein.
5. Overnutrition is a form of malnutrition that can lead to: (a) Kwashiorkor (b) Anaemia (c) Obesity (d) Rickets.

- **Theory Questions:**

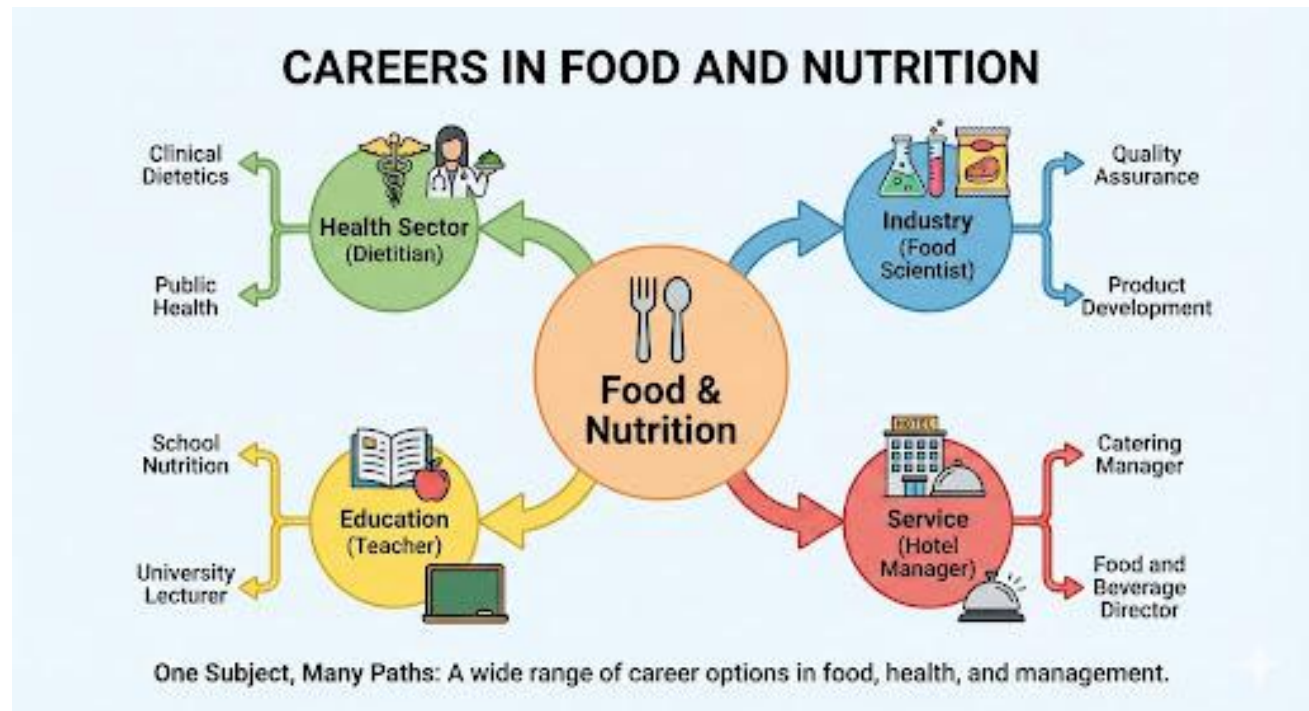
1. Differentiate clearly between "Undernutrition" and "Overnutrition," giving one specific example of a disease for each.
2. List and explain three factors that affect a family's food choices in your locality.
3. What is an antinutrient? Name two examples and a processing method to reduce their effects.

5. Assignment (Week 1)

- Write a three-paragraph "advice column" for a new mother with a limited budget.
 - **Paragraph 1:** Explain the *importance* of good nutrition for her child's growth and health.
 - **Paragraph 2:** Suggest two practical ways she can make better food choices without spending a lot of money (e.g., buying local foods in season, combining plant proteins like rice and beans).
 - **Paragraph 3:** Explain the risk of relying *only* on cheap starchy paps (like *akamu* or *gari* water) to feed her child.

WEEK 2: CAREERS IN FOOD AND NUTRITION

Topic: Career Opportunities



1. Careers in Food and Nutrition

- **Introduction:** Studying Food and Nutrition is a gateway to a wide range of diverse, essential, and stable careers in health, education, industry, and the service sector.

1. Teaching (Education Sector):

- **Role:** Educating students in subjects like Home Economics, Food and Nutrition, or Catering Craft.
- **Workplace:** Primary schools, secondary schools, and tertiary institutions (Colleges of Education, Polytechnics, Universities - as Lecturers).
- **Requires:** NCE, B.Sc.Ed., M.Ed., Ph.D.

2. Dietetics (Health Sector):

- **Role:** A **Dietitian** (or **Clinical Dietitian**) is a highly trained, *legally regulated* health professional who assesses, diagnoses, and treats diet and nutrition-related problems. They are the "food doctors" who can prescribe therapeutic diets.
- **Workplace:** Hospitals (Government & Private), Health Maintenance Organizations (HMOs), primary care clinics, private practice.
- **Jobs:** Prescribing special diets for patients (e.g., low-salt diet for hypertension, renal diet for kidney disease, diabetic diet).

3. Community Nutrition (Public Health Sector):

- **Role:** A **Community Nutritionist** or **Public Health Nutritionist** focuses on the nutritional problems of a *population* (the community, an LGA, a whole state) rather than one-on-one patient care.
- **Workplace:** Government ministries (Health, Agriculture), Local Government Primary Health Centres, international NGOs (e.g., UNICEF, WHO), and non-profits.
- **Jobs:** Running feeding programs (e.g., school feeding), developing public education campaigns (e.g., on breastfeeding, sanitation), and formulating government food policy.

4. Food Science and Technology (Industry Sector):

- **Role:** A **Food Scientist/Technologist** applies chemistry, biology, microbiology, and engineering to the food industry. They are concerned with food from "farm-gate to plate."
- **Workplace:** Food processing companies (e.g., Nestle, Dangote Foods, UAC Foods, Coca-Cola, Nigerian Breweries).
- **Jobs:**
 - **Product Development:** Creating new food products (e.g., a new flavour of Indomie, a new type of yogurt, a shelf-stable juice).
 - **Quality Control (QC) / Quality Assurance (QA):** Testing food to ensure it is safe, high-quality, and meets legal standards (e.g., NAFDAC, SON).
 - **Food Microbiology:** Studying and using microbes (good ones like yeast for bread/beer; bad ones like *Salmonella*).

- **Sensory Analysis:** Running taste panels to test a product's flavour, texture, and appearance.

5. Food Engineering (Industry Sector):

- **Role:** A specialized engineer who designs, installs, operates, and maintains the *machinery, factories, and processes* used to manufacture food on a large scale.
- **Example:** Designing a large-scale flash-fryer for noodles, a bottling plant for Fanta, or an automated packaging system.

6. Hotel and Catering Management (Service Sector):

- **Role:** Manages the "food and beverage" side of the hospitality and service industry.
- **Workplace:** Hotels, restaurants, airlines (flight catering), cruise ships, industrial canteens (for schools or companies), event planning companies.
- **Jobs:**
 - **Food & Beverage Manager:** Manages all food operations in a large hotel.
 - **Catering Manager:** Plans and manages food for large events (weddings, conferences).
 - **Executive Chef:** The head of the kitchen, a culinary artist and manager.

7. Specialized Nutrition Careers (Emerging Fields):

- **Sports Nutritionist:** Works with athletes and teams to optimize performance through diet.
- **Nutrigenetics/Nutrigenomics:** A cutting-edge research field studying the relationship between an individual's genes, nutrition, and health.

2. Evaluation (Week 2)

• Objective Questions:

1. A professional who prescribes therapeutic diets for patients in a hospital is a: (a) Food Scientist (b) Clinical Dietitian (c) Hotel Manager (d) Teacher.
2. The career focused on creating new food products and ensuring safety in a factory is: (a) Dietetics (b) Food Science and Technology (c) Teaching (d) Catering.

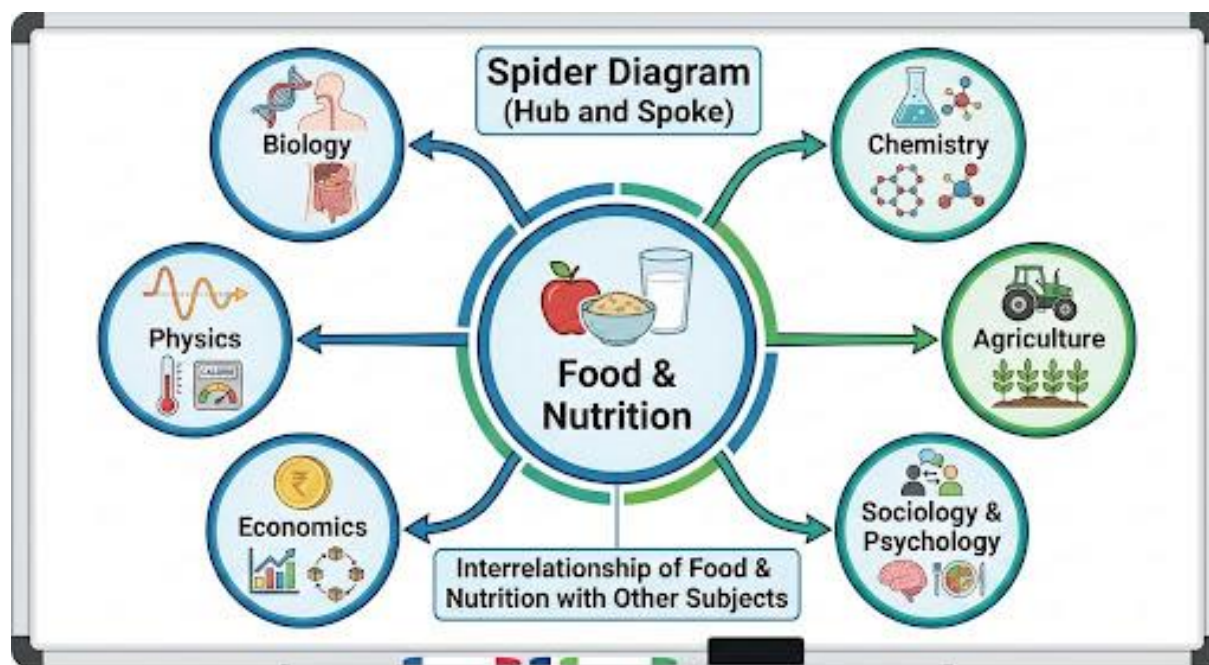
3. NAFDAC (National Agency for Food and Drug Administration and Control) would most likely employ a: (a) Hotel Manager (b) Food Scientist (c) Chef (d) Community Nutritionist.
 4. A person who manages a restaurant in a large hotel is in: (a) Food Engineering (b) Hotel and Catering Management (c) Community Nutrition (d) Clinical Nutrition.
 5. Which career focuses on the nutritional problems of a whole population, like an LGA? (a) Clinical Dietitian (b) Community Nutritionist (c) Food Engineer (d) Executive Chef.
- **Theory Questions:**
 1. What is the main difference between a **Clinical Dietitian** and a **Community Nutritionist**? (Hint: Think about *who* they serve).
 2. Describe two different jobs a **Food Scientist** might do in a company like Nestle or Dangote Foods.

3. Assignment (Week 2)

- Choose *one* career from this week's lesson that you find interesting.
- **Part 1:** Write one paragraph describing why you find it interesting.
- **Part 2:** Go online or ask an adult. Find out *two* subjects (besides Food & Nutrition) that you *must* be good at to succeed in that career (e.g., Chemistry, Biology, Maths, Business Studies, Physics).
- **Part 3:** Write a paragraph explaining *why* those two subjects are essential for that specific job.

WEEK 3: INTERRELATIONSHIP OF FOOD AND NUTRITION WITH OTHER SUBJECTS

Topic: Links to Other Fields of Study



1. Relationship Between Food & Nutrition and Other Subjects

- **Introduction:** Food and Nutrition is not an isolated subject. It is an *applied science* that borrows and integrates principles from many other fields.

1. Biology & Microbiology:

- **Digestion & Absorption:** Understanding the anatomy and physiology of the digestive system (stomach, intestines, enzymes) is pure biology.
- **Metabolism:** How the body's cells use nutrients (e.g., cellular respiration, Krebs's cycle) is a core biological process.
- **Microbiology:** Explains food spoilage (bacteria, molds, yeasts) and food preservation (killing microbes). It also explains *fermentation* (using "good" microbes to create foods like yogurt, *iru*, *fufu*, bread).

- **Food Sources:** Botany (plants) and Zoology (animals) are the study of our food sources.

2. Chemistry (The Foundation):

- **Composition:** All nutrients are *chemicals* with specific structures (e.g., Carbohydrates are $C_n(H_2O)_n$, Proteins are chains of amino acids CHON).
- **Food Reactions:** Chemistry explains all changes during cooking and processing.
 - **Maillard Reaction:** (Browning of meat, toasting of bread).
 - **Caramelization:** (Browning of sugar).
 - **Emulsions:** (How oil and water mix in milk, mayonnaise, *egusi* soup).
 - **pH:** How acidity/alkalinity affects food preservation (e.g., using vinegar) and taste.
- **Food Tests:** All tests for nutrients (e.g., Iodine test for starch, Biuret test for protein) are chemical reactions.

3. Physics:

- **Heat Transfer:** Physics explains the *methods* of cooking.
 - **Conduction:** Heat moves *through* a solid (e.g., frying pan to food).
 - **Convection:** Heat moves *through* liquid or gas (e.g., boiling water, baking in an oven).
 - **Radiation:** Heat moves as waves (e.g., grilling, microwave oven).
- **Food Processing:** Physics explains the principles behind machines like freezers (thermodynamics), blenders (mechanics), and pressure cookers (gas laws).
- **Measurement:** Concepts of mass, weight, and volume are essential for recipes.

4. Agriculture:

- **Food Source:** This is the most obvious link. Agriculture is the *production* of the raw food (crops, livestock) that we study.
- **Food Security:** Agricultural policy and success determine what food is available (food security) and how much it costs.
- **Food Safety:** The use of pesticides, herbicides, and fertilizers in agriculture has a direct impact on the safety of the food we eat.
- **New Varieties:** Agriculture (biotechnology) develops crops with higher nutritional value (biofortification), e.g., Vitamin A-fortified cassava.

5. Economics:

- **Supply and Demand:** Economics explains *why* food prices change (e.g., why tomatoes are cheap in season and expensive off-season).
- **Income & Budgeting:** Economics deals with family income, budgeting, and poverty, which (Week 1) are the main factors affecting food choice and nutritional status.
- **Food Policy:** Government policies on importation, taxation, and subsidies for food are all economic issues.

6. Geography:

- **Climate and Soil:** Geography explains *why* certain foods grow in certain places (e.g., yams and cassava in the warm, wet South; millet and sorghum in the drier North).
- **Food Distribution:** Geography explains the transportation networks (roads, water) and logistics used to move food from farms to cities.

7. Mathematics:

- **Recipes:** Calculating, scaling (doubling, halving), and adjusting ingredient quantities and ratios.
- **Dietary Planning:** Calculating a person's energy (Calorie/Joule) needs, Body Mass Index (BMI), and nutrient percentages for a balanced diet.
- **Budgeting:** Managing food costs and "best buy" calculations.
- **Food Science:** Used for all calculations in a lab (e.g., concentrations, statistical analysis of taste panels).

8. Psychology & Sociology:

- **Psychology:** Explains *why* we eat, not just *what*. (e.g., eating disorders like anorexia, "comfort eating" due to stress, food phobias).
- **Sociology/Anthropology:** Studies food as cultural identity, the social roles of feasting and fasting, and food taboos.

2. Evaluation (Week 3)

- **Objective Questions:**

1. The study of how enzymes break down food in the stomach is an example of the link with: (a) Physics (b) Economics (c) Biology (d) Maths.
2. Boiling water or baking in an oven transfers heat by: (a) Conduction (b) Radiation (c) Convection (d) Metabolism.
3. The subject that explains *why* yams grow in the South and millet in the North is: (a) Physics (b) Geography (c) Chemistry (d) Maths.
4. The Maillard reaction (browning of meat) and caramelization are examples of the link with: (a) Chemistry (b) Economics (c) Agriculture (d) Physics.
5. Understanding food supply, demand, and family budgets connects nutrition to: (a) Physics (b) Geography (c) Economics (d) Biology.

• **Theory Questions:**

1. Explain the interrelationship between Food & Nutrition and *Economics*, using two specific examples.
2. Give two specific examples of how *Physics* is applied in the kitchen.
3. Explain the link between Food & Nutrition and *Microbiology*, giving one positive and one negative example.

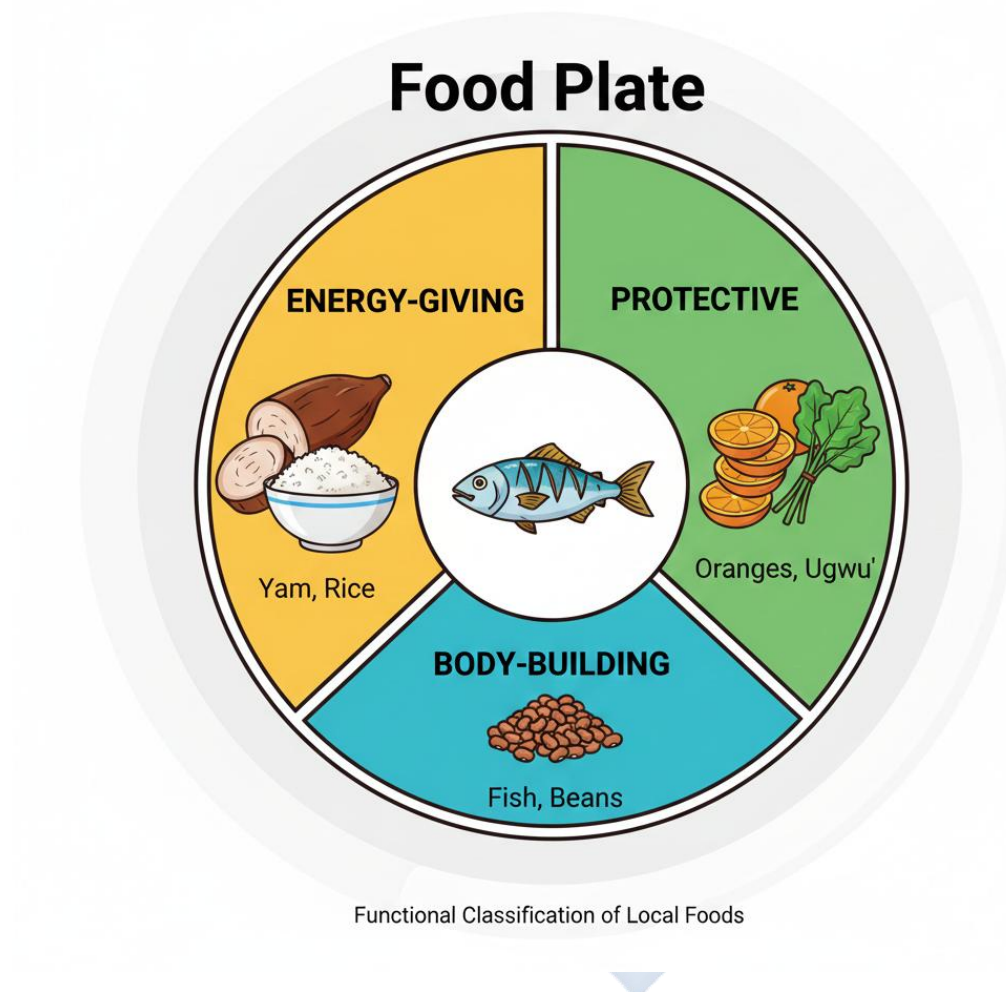
3. Assignment (Week 3)

- Choose *three* different subjects from this week's lesson (e.g., Chemistry, Economics, Geography).
- For *each* subject, write one full paragraph explaining its relationship to Food and Nutrition. You must use specific examples not already used in the theory questions (e.g., for Chemistry, you could discuss *emulsions* or *food tests*).

WEEK 4: FOODS IN THE LOCALITY

Topic: Classifying Local Foods by Nutrients and Functions

FOODS IN THE LOCALITY



1. Foods in the Locality and their Nutrients

- **Introduction:** Nigeria is blessed with a huge variety of staple foods, soups, fruits, and vegetables. We can classify these local foods by the *main* nutrient they provide or by their *main function* in the body.
- **A. Classification by Predominant Nutrient**
 - **1. Foods Rich in Carbohydrates (Energy Foods)**

- **Function:** To provide the body with its main source of energy (fuel) for work, play, and all body processes. Also provides dietary fibre.
- **Local Examples:**
 - **Roots & Tubers:** Yam, Cassava (processed into *gari*, *fufu*, *lafun*, *akpu*, *abacha*), Cocoyam, Sweet Potato, Water Yam.
 - **Grains (Cereals):** Rice (local & imported), Maize (corn), Millet, Sorghum (guinea corn), Acha (fonio).
 - **Processed Cereals:** *Tuwo*, *Ogi/Akamu/Pap*, Bread (from wheat), Spaghetti, Noodles, *Semovita*.
 - **Fruits:** Plantain (ripe & unripe), Banana.
- **2. Foods Rich in Proteins (Body-Building Foods)**
 - **Function:** For growth (especially in children), repairing tissues, making enzymes, hormones, and antibodies.
 - **Local Examples:**
 - **Plant Protein (LBV - Low Biological Value):** Beans (black-eyed, brown), Soya beans (used for *soya-milk*, *awara*), Groundnuts (peanuts), Egusi (melon seeds), Ogbono, Ukwá (breadfruit), *Bambara* nuts.
 - **Animal Protein (HBV - High Biological Value):** Beef (*Suya*), Goat meat (*asun*), Chicken, Turkey, Fish (tilapia, mackerel/titus, catfish, croaker), Crayfish, Snails, Periwinkles (*Isam*), Eggs, Milk, Cheese (*wara*).
- **3. Foods Rich in Fats & Oils (Energy-Reserve Foods)**
 - **Function:** Provide a *concentrated* source of energy, insulate the body (keep warm), protect vital organs, and transport fat-soluble vitamins (A, D, E, K).
 - **Local Examples:**
 - **Oils:** Red Palm Oil, Groundnut Oil, Coconut Oil, Melon Seed (Egusi) Oil, Soya bean Oil.
 - **Fats:** Fat on meat (beef, pork, ram), Butter, Margarine.

- **Other (High-Fat Seeds/Fruits):** Avocado, Ogbono, Egusi, Groundnuts, Coconut.
- **4. Foods Rich in Vitamins & Minerals (Protective & Regulatory Foods)**
 - **Function:** To protect against illness, boost the immune system, regulate all body processes (e.g., vision, blood clotting, bone strength, nerve function).
 - **Local Examples:**
 - **Vegetables (Dark Green Leafy):** *Ugwu* (fluted pumpkin leaf), *Ewedu* (jute leaf), *Tete* (African spinach), *Soko*, Waterleaf, Bitterleaf.
 - **Other Vegetables:** Okra, Peppers, Tomatoes, Onions, Carrots, Cabbage.
 - **Fruits:** Oranges, Mangoes, Pawpaw (papaya), Pineapples, Bananas, Guavas, Watermelon, *Ube* (African pear), *Agbalumo* (African star apple).
 - **Animal (Rich in specific minerals/vitamins):** Liver (Iron, Vit. A), Crayfish/small fish (Calcium).
- **B. Classification by Function (A Simpler Way)**
 - **1. Energy-Giving Foods:** (Primarily Carbohydrates & Fats).
 - *Function:* Provide fuel for the body.
 - *Examples:* Rice, Yam, Gari, Palm Oil, Bread.
 - **2. Body-Building Foods:** (Primarily Proteins).
 - *Function:* For growth and repair.
 - *Examples:* Beans, Fish, Meat, Eggs, Milk, Egusi.
 - **3. Protective Foods:** (Primarily Vitamins & Minerals).
 - *Function:* To protect from illness and regulate the body.
 - *Examples:* *Ugwu*, Oranges, Pawpaw, Tomatoes, Carrots.

2. Evaluation (Week 4)

- **Objective Questions:**

1. Yam, Gari, and Rice are local foods classified as: (a) Protein foods (b) Carbohydrate foods (c) Fat foods (d) Vitamin foods.
2. Which of these is a local source of High Biological Value (HBV) protein? (a) Beans (b) Egusi (c) Fish (d) Ogbono.
3. Ugwu, Ewedu, and Okra are excellent sources of: (a) Carbohydrates (b) Fats (c) Vitamins and Minerals (d) Animal protein.
4. The primary *function* of foods like palm oil and groundnuts is to: (a) Build the body (b) Provide concentrated energy (c) Protect from illness (d) Provide fibre.
5. Tuwo masara, Ogi, and Akpu are all processed from: (a) Roots & Tubers (b) Grains (Cereals) (c) Both A and B (d) Legumes.

• **Theory Questions:**

1. List three local foods that are rich in protein. Classify them as either Plant Source (LBV) or Animal Source (HBV).
2. What is the main *function* of carbohydrate foods? List three local examples (one root, one grain, one fruit).
3. Using the "Classification by Function," list one local food for each group: (a) Energy-Giving, (b) Body-Building, (c) Protective.

3. Assignment (Week 4)

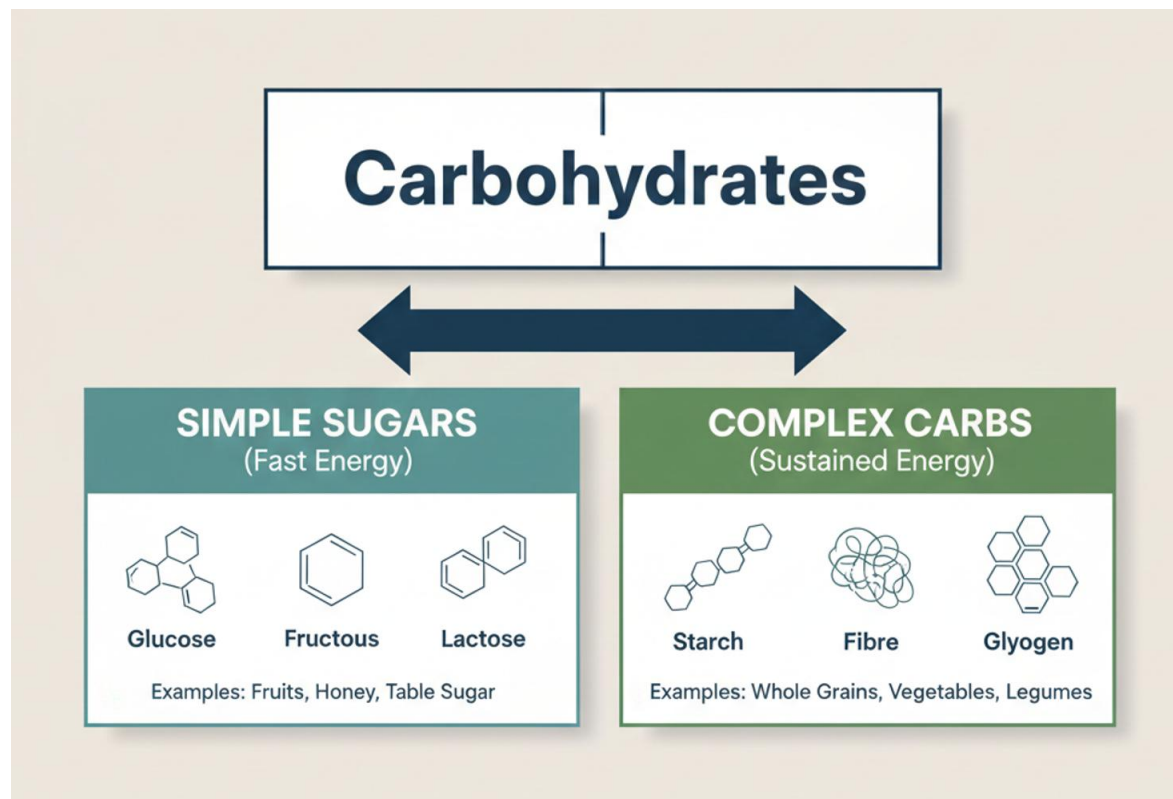
- **Project:** Plan a *one-day* menu (Breakfast, Lunch, and Dinner) for a teenage student using *only* common Nigerian foods.
- **Format:** Create a table with four columns:

Meal	Food Items (e.g., "Yam and Egg Sauce")	Main Nutrient Provided	Main Function
Breakfast			
Lunch			
Dinner			

- **Goal:** Your full-day menu must be *balanced* and include at least one food from each of the three main *functions* (Energy, Building, Protective).

WEEK 6: BASIC FOOD NUTRIENTS (CARBOHYDRATES)

Topic: Carbohydrates



1. Composition, Classification, Functions, Deficiencies, Sources

- **A. Composition:**

- Carbohydrates are organic compounds made of three elements: **Carbon (C)**, **Hydrogen (H)**, and **Oxygen (O)**.
- The general chemical formula is $C_n(H_2O)_n$.
- The Hydrogen and Oxygen are usually in a 2:1 ratio, the same as in water (H_2O).

- **B. Classification:**

- **1. Monosaccharides (Simple Sugars):**

- **Definition:** The simplest "single" sugar units (monomers). They are sweet, dissolve in water, and can be absorbed directly by the body.
- **Glucose ($C_6H_{12}O_6$):** "Blood sugar." The body's main fuel. Found in fruits.

- **Fructose (C₆H₁₂O₆):** "Fruit sugar." The sweetest sugar. Found in honey, fruits.
- **Galactose (C₆H₁₂O₆):** Found in milk, as part of lactose.
- **2. Disaccharides (Double Sugars):**
 - **Definition:** Two monosaccharides joined together (with the loss of one water molecule). They are sweet and soluble.
 - **Sucrose:** (Glucose + Fructose). Table sugar (from sugarcane).
 - **Lactose:** (Glucose + Galactose). "Milk sugar."
 - **Maltose:** (Glucose + Glucose). "Malt sugar." Found in malted grains (e.g., Milo, Maltina) and germinating seeds.
- **3. Polysaccharides (Complex Carbohydrates):**
 - **Definition:** Long chains (polymers) of many sugar units (usually glucose). They are *not* sweet and are *insoluble* in cold water.
 - **Starch:** The storage form of energy in *plants*. (e.g., yam, cassava, rice, potato). Must be cooked to be digested.
 - **Glycogen:** The storage form of energy in *animals*. Stored in the liver and muscles as a ready-fuel reserve.
 - **Cellulose (Dietary Fibre):** The structural part of plant cell walls. *Humans cannot digest it* because we lack the *cellulase* enzyme.
 - **Pectin:** A polysaccharide found in fruits (like apples, oranges) that helps jam to set.
- **C. Functions:**
 1. **Main Source of Energy:** This is the #1 function. The body "burns" carbohydrates (glucose) for fuel for all activities (physical, mental, metabolic). 1 gram provides ~4 kcal (17 kJ) of energy.
 2. **Protein-Sparing Function:** This is a critical concept. When the diet has *enough* carbohydrates for energy, it "spares" the protein to be used for its main job (growth and repair). If there are *not enough* carbs, the body will be forced to burn precious protein for energy, which is a waste and leads to muscle loss.
 3. **Provides Dietary Fibre (Cellulose):**

- Fibre adds bulk (roughage) to food.
- It prevents **constipation** by absorbing water and helping waste (faeces) move through the intestines (peristalsis).
- It helps you feel "full" (satiety), which can help with weight control.
- It acts as a **prebiotic** (food for "good" gut bacteria).
- 4. **Adds Sweetness and Flavour:** Sugars (mono/disaccharides) make food palatable.
- 5. **Fuel for the Brain:** The brain's preferred and primary fuel source is glucose.
- **D. Dietary Deficiencies / Excess:**
 - **Deficiency:**
 - **Lack of Energy:** Tiredness, weakness, fatigue.
 - **Weight Loss:** The body burns its own fat and muscle (protein) for fuel.
 - **Protein Wasting:** Protein is used for energy instead of repair (see Function #2).
 - **Ketosis:** A serious condition where the body, lacking carbs, burns too much fat. This produces acidic byproducts called *ketones*, which build up in the blood.
 - **Excess:**
 - The body converts excess carbohydrates (that aren't used for energy or stored as glycogen) into **fat** (adipose tissue) for long-term storage.
 - This leads to **weight gain** and **obesity**.
 - Excess *sugar* intake leads to **tooth decay (dental caries)**.
 - Chronic high sugar/carb intake is linked to **Type 2 Diabetes** (insulin resistance).
- **E. Food Sources:**
 - **Starch:** Yam, gari, fufu, rice, maize, bread, potato, plantain, pasta, *semovita*.
 - **Sugar:** Table sugar, honey, fruits, soft drinks, cakes, biscuits, sweets.
 - **Fibre:** *Ewedu*, okra, whole grains (e.g., local rice), beans, fruits, and vegetables.

2. Evaluation (Week 6)

- **Objective Questions:**

1. Carbohydrates are composed of: (a) Carbon, Hydrogen, Nitrogen (b) Carbon, Hydrogen, Oxygen (c) Carbon, Nitrogen, Oxygen (d) Carbon, Sulphur, Hydrogen.
2. Which of these is a monosaccharide? (a) Starch (b) Lactose (c) Sucrose (d) Glucose.
3. The main function of carbohydrates is: (a) Growth (b) Energy (c) Repair (d) Protection.
4. The indigestible part of plants that prevents constipation is: (a) Starch (b) Glycogen (c) Cellulose (Fibre) (d) Glucose.
5. The storage form of carbohydrate in the human liver and muscles is: (a) Starch (b) Glycogen (c) Cellulose (d) Glucose.

• **Theory Questions:**

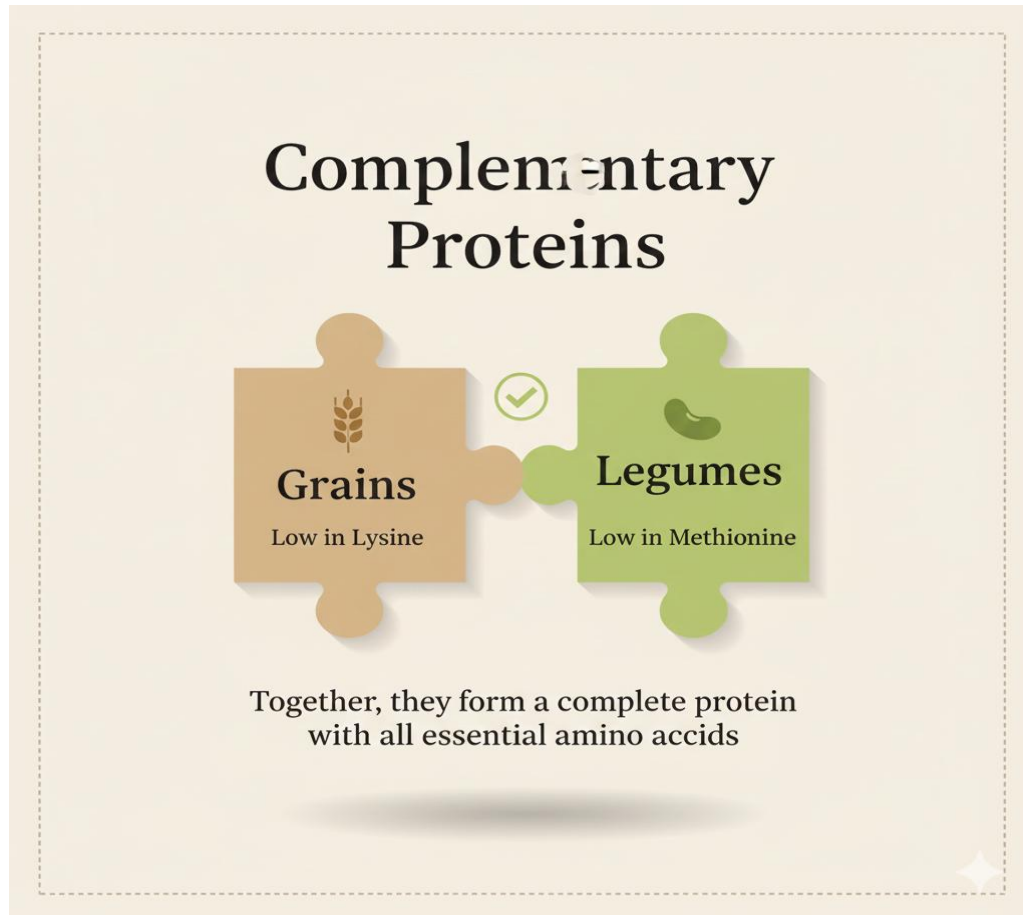
1. What is the "protein-sparing function" of carbohydrates? Explain what happens if the diet is *deficient* in carbohydrates.
2. List the three main *classes* of carbohydrates. For each class, name one specific *example* (e.g., Fructose) and one *food source*.

3. Assignment (Week 6)

- Explain in two paragraphs why a manual labourer (e.g., a bricklayer) and an office worker (e.g., a secretary) have *different* carbohydrate needs.
 - **Paragraph 1:** Who needs more carbohydrates and *why*? (Link to the *function* of carbs).
 - **Paragraph 2:** What are the *consequences* for *each* person if their carbohydrate intake is *excessive* for their lifestyle?

WEEK 8: PROTEINS

Topic: Proteins



Composition, Classification, Functions, Deficiencies, Sources

- **A. Composition:**

- Proteins are very large, complex organic compounds made of: **Carbon (C)**, **Hydrogen (H)**, **Oxygen (O)**, and **Nitrogen (N)**. (Some also contain Sulphur).
- The presence of **Nitrogen** is what distinguishes protein from carbohydrates and fats.
- The basic "building blocks" of all proteins are **Amino Acids**.
- Amino acids are joined together by **peptide bonds** to form long chains (polypeptides).

- There are over 20 amino acids. 9 of them are **Essential Amino Acids (EAAs)** (e.g., Lysine, Methionine, Tryptophan). The body *cannot* make these; they *must* come from food. 11 are **Non-Essential** (the body can make them).
- **B. Classification (By Quality):**
 - **1. High Biological Value (HBV) Proteins:**
 - **Definition:** "Complete" proteins. They contain *all 9* essential amino acids in the correct proportions needed by the human body for growth.
 - **Source:** All **animal** sources.
 - **Examples:** Meat, fish, poultry (chicken), eggs, milk, cheese (*wara*).
 - **2. Low Biological Value (LBV) Proteins:**
 - **Definition:** "Incomplete" proteins. They are low in or *lack* one or more of the essential amino acids. The missing one is called the "**limiting amino acid**."
 - **Source:** All **plant** sources.
 - **Examples:** Beans, groundnuts, grains (rice, wheat), egusi, ogbono.
 - *Grains* are limited by *Lysine*.
 - *Legumes (Beans)* are limited by *Methionine*.
 - **Exception:** Soya bean is a plant protein but is *almost* HBV (a very high-quality plant protein).
- **C. Complementary and Supplementary Values of Protein**
 - This is how we get high-quality protein from cheaper plant sources.
 - **Complementary Protein:**
 - **Definition:** Combining *two or more LBV proteins* in the same meal so that their amino acids compensate for each other's weaknesses (i.e., one provides the limiting amino acid of the other).
 - **The Classic Example: Rice and Beans.**
 - Rice (grain) lacks Lysine.
 - Beans (legume) lacks Methionine.
 - Eaten together, they provide *all 9* EAAs and form a "complete" HBV protein meal.

- **Other Examples:** Bread (grain) and Peanut Butter (legume); *Eba* (cassava) and *Egusi* soup (seed).
- **Supplementary Protein:**
 - **Definition:** Adding a *small amount of HBV protein* to a larger LBV protein meal to improve its overall quality.
 - **Examples:**
 - Adding **crayfish** or **fish** to *Gari* or Vegetable Soup.
 - Adding a little **egusi** to *Ewedu* soup.
 - Eating **bread** (LBV) with an **egg** (HBV).
- **D. Functions:**
 1. **Growth:** This is the #1 primary function. Essential for building new cells, muscles, and tissues. This is vital for infants, children, teenagers, and pregnant women.
 2. **Repair:** Replaces and repairs all worn-out or damaged body tissues (e.g., healing a cut, replacing red blood cells).
 3. **Synthesis of Body Compounds:** Proteins are needed to make vital functional compounds:
 - **Enzymes:** (e.g., *Amylase*, *Pepsin*) that digest food.
 - **Hormones:** (e.g., *Insulin*) that regulate body processes.
 - **Antibodies:** That form the immune system and fight infection.
 4. **Fluid Balance:** Proteins (like albumin) in the blood attract water, which helps to keep fluid in the right place (in blood vessels, not tissues).
 5. **Secondary Energy:** The body can use protein for energy (1 gram = 4 kcal) *only if* there are not enough carbohydrates or fats. This is a *wasteful* use of protein and is the body's last resort.
- **E. Dietary Deficiencies (Protein-Energy Malnutrition - PEM):**
 - **Kwashiorkor:** (The "disease the first child gets when the second is born").
 - **Cause:** Severe *protein* deficiency (but with *some* carb calories, e.g., from pap).

- **Symptoms:** Swollen belly and feet (**oedema** - due to lack of protein for fluid balance), changes in skin/hair pigment (reddish hair), poor growth, misery, fatty liver.
- **Marasmus:** (From the Greek "to waste away").
 - **Cause:** Severe *protein AND energy (calorie)* deficiency. It is total starvation.
 - **Symptoms:** Severe wasting ("skin and bones"), "old man" face, severe weight loss, no oedema.
- **F. Food Sources:**
 - **Animal (HBV):** Meat, fish, eggs, milk, chicken, crayfish, snails, cheese (*wara*).
 - **Plant (LBV):** Beans, groundnuts, lentils, egusi, ogbono, soya beans, ukwa, *Bambara* nuts.

2. Evaluation (Week 8)

- **Objective Questions:**
 1. Proteins are made of C, H, O, and: (a) Iron (b) Calcium (c) Nitrogen (d) Iodine.
 2. Which of these is a High Biological Value (HBV) protein? (a) Beans (b) Rice (c) Eggs (d) Groundnuts.
 3. Eating "Rice and Beans" is an example of: (a) Supplementary protein (b) Complementary protein (c) Marasmus (d) Kwashiorkor.
 4. A disease caused by *only* protein deficiency (with enough calories) is: (a) Marasmus (b) Kwashiorkor (c) Scurvy (d) Rickets.
 5. The basic "building blocks" of proteins are called: (a) Fatty acids (b) Amino acids (c) Glucose (d) Vitamins.
- **Theory Questions:**
 1. What is the difference between an HBV protein and an LBV protein? Name the *type of food source* for each.
 2. Explain "supplementary protein" and give two local examples.
 3. Why does a child with Kwashiorkor have a swollen belly (oedema)? (Hint: Link to the *functions* of protein).

3. Assignment (Week 8)

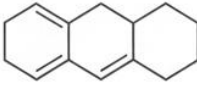
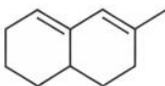
- Plan a simple, cheap, and nutritious lunch for a 2-year-old child using the principle of **complementary proteins**.
 - **Part 1:** Name the meal (e.g., "Mashed Rice and Beans").
 - **Part 2:** List the two main LBV ingredients.
 - **Part 3:** Explain in one paragraph *why* this combination is a complete protein, mentioning the "limiting amino acids" (Lysine and Methionine) if you can.



WEEK 9: FATS AND OILS (LIPIDS)

Topic: Fats and Oils

FATS & OILS

		OILS
STATE AT ROOM TEMP.	 Solid	 Liquid
SOURCE	 Animal / Tropical Plant	 Plant / Fish
CHEMICAL TYPE		Unsaturated
Saturated		
EXAMPLES	Butter, Lard Coconut Oil	Groundnut Oil, Olive Oil, Fish Oil

1. Differences, Classification, Functions, Deficiencies, Sources

- A. Composition:

- Fats and Oils (Lipids) are also made of **Carbon (C)**, **Hydrogen (H)**, and **Oxygen (O)**, but with *much less* Oxygen than carbohydrates.
- The building blocks of fats are **Glycerol** and **Fatty Acids**. (One unit of fat = 1 Glycerol + 3 Fatty Acids, forming a *Triglyceride*).
- **B. Differences Between Fats and Oils:** | Feature | Fats | Oils | | :--- | :--- | :--- | | **State at Room Temp** | Solid | Liquid | | **Primary Source** | Mostly Animal | Mostly Plant (and Fish) | | **Chemical Type** | Mostly **Saturated** | Mostly **Unsaturated** | | **Examples** | Butter, Lard, Meat fat | Groundnut oil, Olive oil |
 - **Exceptions:** Palm oil and Coconut oil are *plant* sources but are *solid/semi-solid* at room temp and are high in *saturated* fat.
- **C. Classification (By Fatty Acid Structure):**
 - **1. Saturated Fats:**
 - **Description:** "Bad fats." Have *no double bonds* in their chemical structure (they are "saturated" with hydrogen).
 - **Effect:** They tend to *raise bad* blood cholesterol (LDL), increasing the risk of heart disease.
 - **Sources:** Animal fat (on meat), butter, cheese, **Red Palm Oil, Coconut Oil**, Lard.
 - **2. Unsaturated Fats (Mono- & Poly-)**
 - **Description:** "Good fats." Have *one or more double bonds*. They are liquid at room temp.
 - **Effect:** They can help *lower* bad cholesterol and are heart-healthy.
 - **Monounsaturated (one double bond):** Groundnut oil, Olive oil, Avocado.
 - **Polyunsaturated (many double bonds):** Soya bean oil, Fish oil (**Omega-3**), Sunflower oil.
 - **3. Trans-Fats (Hydrogenated Fats):**
 - **Description:** Man-made "very bad fats." Created by an industrial process (hydrogenation) that turns liquid oil into solid fat.
 - **Effect:** Raise bad cholesterol (LDL) AND lower good cholesterol (HDL).
 - **Sources:** Stick margarine, some processed biscuits, pastries, and fast food.

- **D. Functions:**

1. **Concentrated Energy:** Provides the *most* energy per gram (1 gram = 9 kcal / 37 kJ). This is more than *twice* the energy of carbs or protein. It is the body's "reserve" fuel.
2. **Insulation:** A layer of fat under the skin (adipose tissue) acts like a blanket, keeping the body warm.
3. **Protection:** Fat pads (adipose tissue) surround and cushion vital organs (like the heart and kidneys) from physical shock.
4. **Transport:** Acts as a carrier for **Fat-Soluble Vitamins (A, D, E, K)** and helps the body absorb them from the intestine.
5. **Flavour and Satiety:** Fats make food taste better and help you feel "full" (satiety) for longer after a meal.

- **E. Dietary Deficiency / Excess:**

- **Deficiency:** (Rare in Nigeria, but...)
 - Dry, flaky skin and hair loss.
 - Constant feeling of cold (lack of insulation).
 - Deficiency in vitamins A, D, E, K (since they can't be absorbed).
- **Excess:**
 - **Obesity:** Because fats are so energy-dense, excess intake leads to significant weight gain.
 - **Heart Disease:** High intake of *saturated* and *trans* fats leads to high blood cholesterol (LDL), which forms *plaque* (Atherosclerosis) that blocks arteries, causing **hypertension** and **heart attacks**.

- **F. Food Sources:**

- **Animal:** Fat on meat, poultry skin, fish (mackerel, titus), butter, *wara*, egg yolk.
- **Plant:** Red palm oil, groundnut oil, soya bean oil, coconut oil, avocado, egusi, ogbono, groundnuts, margarine.

2. Evaluation (Week 9)

- **Objective Questions:**

1. Fats are different from oils because fats are: (a) Liquid at room temp (b) Solid at room temp (c) From plants (d) Unsaturated.
2. Which of these is a "good" unsaturated fat? (a) Fat on beef (b) Butter (c) Groundnut oil (d) Palm oil.
3. A key function of fat is to: (a) Build muscle (b) Provide immediate energy (c) Transport vitamins A, D, E, K (d) Heal wounds.
4. Excess intake of saturated fats is linked to: (a) Kwashiorkor (b) Scurvy (c) Heart disease (d) Rickets.
5. The building blocks of fats are glycerol and: (a) Amino acids (b) Glucose (c) Fatty acids (d) Starch.

- **Theory Questions:**

1. List two main differences between fats and oils.
2. Explain two functions of fat in the body *other* than providing energy.
3. What is the difference between a saturated fat and an unsaturated fat? Give one food source for each.

Assignment (Week 9)

- **Part 1:** List three common foods your family eats that are high in *saturated* fat.
- **Part 2:** List three common foods your family eats that are high in *unsaturated* fat.
- **Part 3:** Write one paragraph explaining why it is healthier to cook with groundnut oil or soya bean oil than with palm oil (even though palm oil is a good source of pro-Vitamin A).

WEEK 10: VITAMINS & MINERALS

Topic: Vitamins and Minerals

VITAMINS & MINERALS: DEFICIENCY DISEASE QUICK-GUIDE

DEFICIENCY DISEASE		LACKING NUTRIENT	
Scurvy		Vitamin C / Calcium	
Rickets		Vitamin D / Calcium	
Night Blindness		Vitamin A (Thiamine)	
Beriberi B1 (Thiamine)		Pellagra	
Pellagra		Iron / Vitamin B12 / Folate	
Anemia		Iodine	
Osteoporosis		Calcium / Calcium D	

1. Vitamins (Organic)

- **Definition:** Organic (carbon-containing) compounds needed in *tiny* amounts (micronutrients) to regulate body processes. They mainly act as **co-enzymes** ("helpers" for enzymes).
- **A. Fat-Soluble Vitamins (Stored in body fat/liver; can be toxic in excess)**
 - **Vitamin A (Retinol / Beta-carotene):**
 - Functions: Helps with vision in dim light (prevents night blindness), healthy skin, strong immune system.

- Deficiency: **Night blindness**, Xerophthalmia (dry eyes), poor growth.
- Sources: (Retinol) Liver, eggs, milk. (Pro-vitamin Beta-carotene) **Red Palm Oil**, carrots, *ugwu*, mango, pawpaw.
- **Vitamin D (Calciferol - The "Sunshine Vitamin"):**
 - Functions: Helps the body *absorb* **Calcium** and Phosphorus for strong bones and teeth.
 - Deficiency: **Rickets** in children (soft, bent bones), Osteomalacia in adults.
 - Sources: **Sunlight on skin (main source)**, cod liver oil, eggs, fortified milk.
- **Vitamin E (Tocopherol):**
 - Functions: A powerful **antioxidant** (protects cells, especially fats, from damage by free radicals).
 - Sources: Vegetable oils, nuts, seeds, leafy greens.
- **Vitamin K (Phylloquinone):**
 - Functions: Essential for **blood clotting** (activates clotting factors).
 - Deficiency: Blood fails to clot; excessive bleeding.
 - Sources: Leafy green vegetables (*ugwu*, *tete*), liver. (Also made by gut bacteria).
- **B. Water-Soluble Vitamins (NOT stored; excess is passed in urine; needed daily)**
 - **Vitamin B-Complex** (Group of vitamins for energy release and cell metabolism)
 - **B1 (Thiamine):** Releases energy from carbs. Deficiency -> **Beri-beri** (nerve damage, heart failure). (Sources: Whole grains, beans, meat).
 - **B2 (Riboflavin):** Energy release, healthy skin. Deficiency -> Cracks at corners of mouth (*Ariboflavinosis*). (Sources: Milk, eggs, liver).
 - **B3 (Niacin):** Energy release. Deficiency -> **Pellagra** (The 4 Ds: Diarrhoea, Dermatitis, Dementia, Death). (Sources: Meat, fish, groundnuts, tryptophan).
 - **B9 (Folate/Folic Acid):** Vital for making new cells and DNA. Deficiency -> **Anaemia** (large red blood cells). Critical for pregnant women to prevent **neural tube defects** (e.g., *spina bifida*) in the baby. (Sources: Leafy greens, liver, beans).

- **B12 (Cobalamin):** Helps make new cells, healthy nerves. Deficiency -> **Pernicious Anaemia**. (Sources: **Only animal foods** - meat, fish, eggs, milk).
- **Vitamin C (Ascorbic Acid):**
 - Functions: Boosts immune system, heals wounds (makes collagen), helps absorb *non-heme iron*. A powerful antioxidant.
 - Deficiency: **Scurvy** (bleeding gums, lost teeth, weak wounds, joint pain).
 - Sources: **Citrus fruits (oranges, lemons)**, guava, pawpaw, peppers, tomatoes. (Easily destroyed by heat/cooking).

Minerals (Inorganic)

- **Definition:** Inorganic elements (from the earth) needed by the body.
- **A. Macro-Minerals (Needed in larger amounts)**
 - **Calcium (Ca):**
 - Functions: 99% is for strong bones and teeth. 1% is for muscle contraction, nerve function, blood clotting.
 - Deficiency: **Rickets** (with Vit. D), **Osteoporosis** (brittle bones in old age), muscle cramps.
 - Sources: Milk, *wara*, cheese, **crayfish, tiny fish (eaten with bones)**, *ugwu*.
 - **Phosphorus (P):** Works with calcium for bones and teeth. (Sources: Milk, meat, fish).
 - **Sodium (Na) & Potassium (K):** Work together for fluid balance and nerve function.
 - Sodium (Na): (Source: Salt. *Excess* causes high blood pressure).
 - Potassium (K): (Source: Bananas, oranges, vegetables. *Good* for blood pressure).
- **B. Micro-Minerals (Trace Elements - needed in tiny amounts)**
 - **Iron (Fe):**
 - Functions: The most important part of **haemoglobin** in red blood cells, which *carries oxygen* from the lungs to the body.

- Deficiency: **Iron-Deficiency Anaemia** (tired, weak, pale, dizzy, out of breath).
- Sources:
 - **Heme-Iron** (easily absorbed): Liver, red meat.
 - **Non-Heme-Iron** (poorly absorbed): *Ugwu, ewedu*, beans, eggs. (Eating with Vitamin C, like tomato, helps absorption).
- **Iodine (I):**
 - Functions: Needed by the thyroid gland (in the neck) to make hormones that control growth and metabolism.
 - Deficiency: **Goitre** (a large swelling of the thyroid gland in the neck). In infants, can cause **Cretinism** (severe, permanent mental and physical disability).
 - Sources: **Iodized salt** (most important source), seafood, fish.
- **Fluorine (F):** Helps harden tooth enamel, prevents tooth decay. (Source: Fluoridated water, toothpaste).

Evaluation (Week 10)

• Objective Questions:

1. Vitamins A, D, E, and K are known as: (a) Water-soluble (b) Fat-soluble (c) Antinutrients (d) Pro-vitamins.
2. A deficiency of Vitamin C causes: (a) Rickets (b) Goitre (c) Scurvy (d) Beri-beri.
3. A deficiency of Iron causes: (a) Anaemia (b) Goitre (c) Rickets (d) Scurvy.
4. The mineral needed for strong bones (with Vitamin D) is: (a) Iron (b) Iodine (c) Calcium (d) Sodium.
5. Goitre is a swelling in the neck caused by a lack of: (a) Iron (b) Iodine (c) Calcium (d) Vitamin A.
6. A pregnant woman needs extra Folic Acid (Vit. B9) to prevent: (a) Pellagra (b) Scurvy (c) Night blindness (d) Neural tube defects.

• Theory Questions:

1. Why is Vitamin A important, and what are two good *local* sources?

2. Explain the function of Iron and the function of Iodine. What is the *main* deficiency disease for each?
3. What is the difference between Heme-Iron and Non-Heme-Iron? What vitamin helps us absorb non-heme iron?

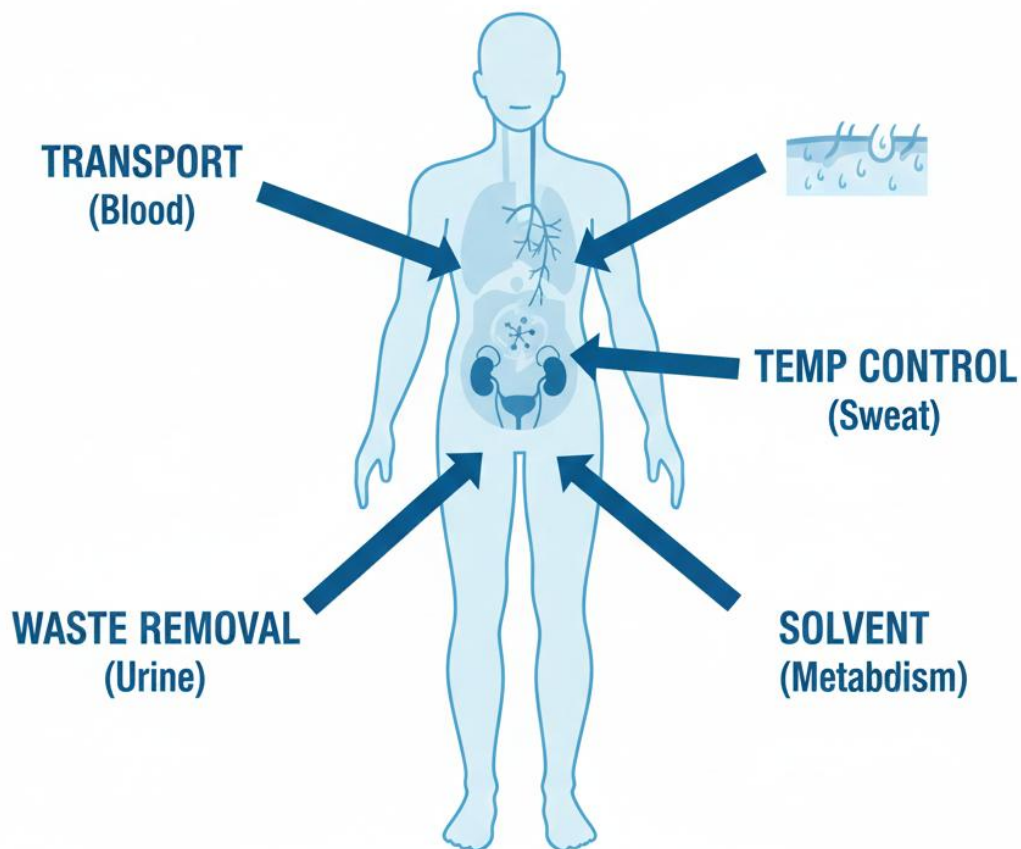
Assignment (Week 10)

- Create a simple table for the following deficiency diseases. | Disease | Nutrient Lacking | One Main Symptom | One Food Source to Prevent It | | :--- | :--- | :--- | :--- | | Kwashiorkor | Protein | Swollen belly | Eggs | | Scurvy | *Vitamin C* | *Bleeding gums* | *Orange/Guava* | | Anaemia (Iron) | *Iron* | *Tiredness/Fatigue* | *Liver / Ugwu* | | Rickets | *Vitamin D / Calcium* | *Bowed (bent) legs* | *Sunlight / Crayfish* | | Goitre | *Iodine* | *Swollen neck* | *Iodized Salt / Seafood* | | Pellagra | *Niacin (B3)* | *Skin rash (Dermatitis)* | *Groundnuts / Meat* | | Night Blindness | *Vitamin A* | *Poor vision in dim light* | *Red Palm Oil / Carrots* |
- Fill in the blank spaces in the table. (I have filled in the first row as an example and the rest as the solution).



THE MULTIMACITED ROLES OF WATER IN THE HUMAN BODY

SUMMARY: WATER IS VITAL FOR TRANSPORT, REGULATION, AND CHEMICAL PROCESSES.



Chemical Composition, Functions, Food Sources

- **A. Chemical Composition:**
 - Water is a simple inorganic chemical compound.
 - Its chemical formula is **H₂O**.
 - It is made of two atoms of **Hydrogen (H)** and one atom of **Oxygen (O)**.

- **B. Functions:**

- Water is the *most essential* nutrient for survival; you can only survive a few days without it. It makes up ~60-70% of the human body.
- 2. **A Universal Solvent:** Water dissolves most other nutrients (sugars, salts, water-soluble vitamins, minerals), waste products (urea), and gases. This allows chemical reactions (metabolism) to occur.
- 3. **Transport Medium:** It is the main component of **blood** (plasma), which transports oxygen, nutrients, and hormones *to* cells and carries CO₂ and waste products *away* from cells.
- 4. **Temperature Regulation (Cooling):** This is a critical function.
 - Water has a *high specific heat capacity* (it takes a lot of energy to heat it up), which prevents the body from overheating.
 - The body *cools* itself by sweating. The *evaporation* of sweat (water) from the skin requires energy (heat), which it takes from the body, thus cooling it down.
- 5. **Digestion and Absorption:**
 - It is the main component of saliva (lubricates food) and digestive juices.
 - It breaks down food by **hydrolysis**.
 - It is the medium in which nutrients are absorbed into the blood.
- 6. **Excretion of Waste:** Water is essential to flush toxic waste products from the body in the form of **urine** (removes urea) and faeces (it softens stool, preventing constipation).
- 7. **Lubrication and Protection:** It acts as a lubricant and cushion for joints (synovial fluid), eyes, and the digestive tract (mucus). It also protects the brain and spinal cord (cerebrospinal fluid).

- **C. Food Sources:**

- 0. **Drinking Water & Beverages:** The main source. This includes plain water, juice, milk, tea, soft drinks, *zobo*.
- 1. **Fruits:** Especially **watermelon**, pawpaw, oranges, pineapple, *agbalumo*.
- 2. **Vegetables:** Especially **cucumber**, lettuce, tomatoes, waterleaf.

3. **Cooked Foods: Soups and Stews** have a very high water content. Pap (*akamu*) is mostly water.

- **D. Water Balance (Intake vs. Output):**

- **Intake:** Must equal output to stay healthy. We get water from:
 1. Drinks
 2. Water in solid food
 3. Metabolic water (made inside our cells).
- **Output:** We lose water through:
 0. **Urine (kidneys):** The main way we control water balance.
 1. **Sweat (skin):** Increases *greatly* with heat and exercise.
 2. **Breath (lungs):** We lose water vapour every time we exhale.
 3. **Faeces (intestines):** (Loss increases with diarrhoea).
- **Dehydration:** Occurs when *output is greater than intake*.
 - **Symptoms:** Thirst, dark-coloured urine, dry mouth, dizziness, confusion. Very dangerous, especially in infants with diarrhoea.

Evaluation (Week 11)

- **Objective Questions:**

1. The chemical formula for water is: (a) CO₂ (b) H₂O (c) CHO (d) CHON.
2. Water is made of two atoms of Hydrogen and one atom of: (a) Oxygen (b) Nitrogen (c) Carbon (d) Calcium.
3. The *main* way the body cools itself down in a hot environment is through: (a) Urination (b) Breathing (c) Sweating and evaporation (d) Drinking.
4. Water carries nutrients and oxygen around the body as the main part of: (a) Sweat (b) Blood plasma (c) Bones (d) Adipose tissue.
5. A condition where water *output* is greater than water *intake* is called: (a) Hydrolysis (b) Dehydration (c) Metabolism (d) Transport.

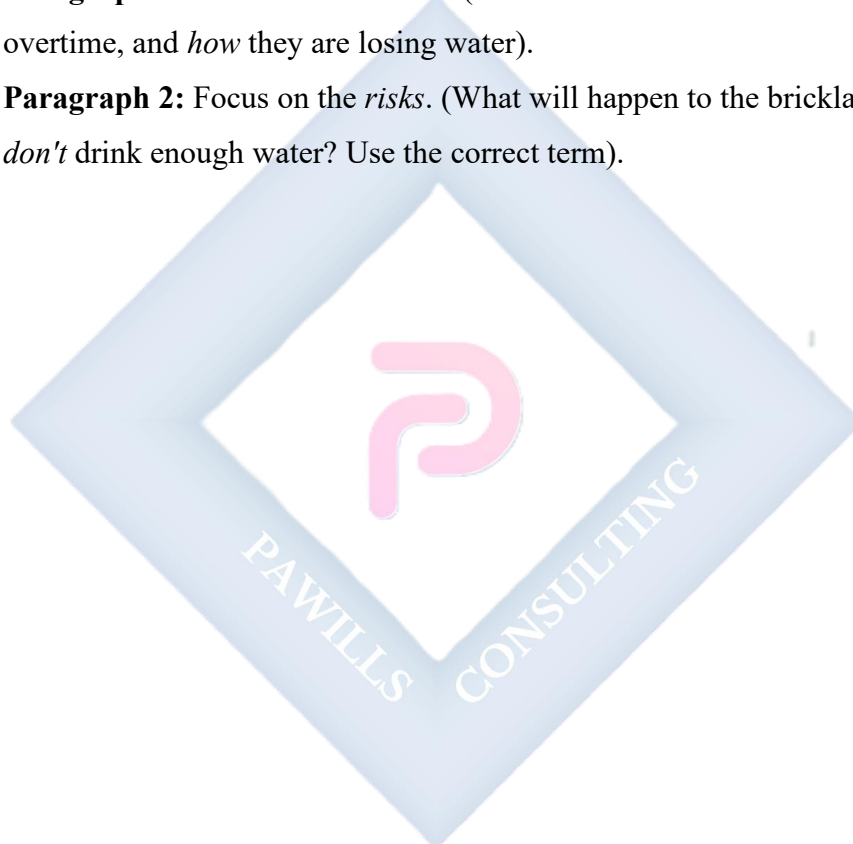
- **Theory Questions:**

1. List three functions of water in the body *other* than digestion.

2. List three *sources* of water in the diet, *other* than plain drinking water.
3. Explain *how* water regulates body temperature.

Assignment (Week 11)

- Explain in detail (two paragraphs) why a person doing hard manual labour (e.g., a bricklayer) in the hot sun needs to drink *much more* water than an office worker in an air-conditioned room.
 - **Paragraph 1:** Focus on water loss. (Mention *which* function of water is working overtime, and *how* they are losing water).
 - **Paragraph 2:** Focus on the *risks*. (What will happen to the bricklayer if they *don't* drink enough water? Use the correct term).



WEEK 12:
REVISION

Plant Nutrition: Essential Nutrient Cheat Sheet

1. Macronutrients: N-P-K  N ₂ K _K   Function: Growth leaves Deficiency: Stunted growth Sources: Fertilizer Compost	2. Calcium (Ca)  Ca Function: Cell Walls, Strength Deficiency: Chlorosis, Curled leaves Sources: Lime, Eggshells	3. Magnesium (Mg) Mg  Function: Blossom-end rot, Curled leaves Deficiency: Chlorosis, Purple spots Sources: Epsom salts, Dolomite
4. Sulfur  S ₂  Functions: Protein synthesis, Enzymes Deficiency: Pale new leaves, Stunted Sources: Gypsum, Manure	5. Micronutrients (Fe, Mn...) Fe B, B, Mn, Zn... Function: Intervascular chlorosis, Potted plants, Trace functions Deficiency: Various localized symptoms Sources: Foliar sprays, Chelates	6. Soil pH Influence Concept: (4-6) Acidic (6-7) Alkaline (7-8)  Impact: Neutral (6-7) Impact: Nutrient availability Key: Most nutrients prefer slightly acidic to neutral pH for optimal uptake

A Condensed, At-A-Glance Study Aid for Healthy Plants

- **Topic:** Review of all topics covered in the term.
- This week is for reviewing all key concepts from Week 1 to Week 11 in preparation for the terminal examination.
- **Summary of Key Concepts:**

- **Week 1:** Definitions (Food, Nutrition, Malnutrition, Antinutrient, Pro-vitamin), Importance of food (Energy, Growth, Protection, Social), Factors affecting choice (Income, Season, Culture, Education, Time).
- **Week 2:** Careers (Teaching, Clinical Dietitian, Community Nutritionist, Food Scientist, Catering, Food Engineer, Sports Nutritionist).
- **Week 3:** Interrelationship (Biology-Digestion/Microbes, Chemistry-Nutrients/Reactions, Physics-Heat/Machines, Econ-Price/Policy, Agri-Source/Safety, Geog-Climate).
- **Week 4:** Local Foods & Functions (Energy: Yam, Gari; Body-Building: Beans, Fish; Protective: *Ugwu*, Orange).
- **Week 6 (Carbs):** CHO. 3 types (Mono-Glucose, Di-Sucrose, Poly-Starch/Fibre). Main function: **Energy**. "Protein-sparing" function. Deficiency: Tiredness, Ketosis. Excess: Obesity. Sources: Rice, Yam, Bread.
- **Week 8 (Protein):** CHON (Amino Acids). 2 types (HBV-Animal/Eggs, LBV-Plant/Beans). Main function: **Growth & Repair**. Deficiency: **Kwashiorkor, Marasmus**. (Complementary: Rice + Beans).
- **Week 9 (Fats):** CHO (Fatty Acids). Solid vs. Liquid. 3 types (Saturated-Bad/Palm Oil, Unsaturated-Good/Groundnut Oil, Trans-Very Bad/Margarine). Main function: **Energy Reserve, Transport Vit. A, D, E, K**. Excess: Obesity, Heart Disease.
- **Week 10 (Vitamins/Minerals):** (Micronutrients)
 - Vitamins (Organic, Co-enzymes):
 - Fat-Soluble: A (Night blindness), D (Rickets), E (Antioxidant), K (Clotting).
 - Water-Soluble: B1 (Beri-beri), B3 (Pellagra), B9 (Anaemia, Birth Defects), B12 (Anaemia), C (Scurvy).
 - Minerals (Inorganic):
 - Macro: Calcium (Rickets, bones).
 - Micro: Iron (Anaemia), Iodine (Goitre).
- **Week 11 (Water):** H₂O. Functions: Transport (blood), Temperature (sweat), Solvent, Lubricant, Excretion (urine). Deficiency: Dehydration.

- **Assignment (Week 12):**

- Create a "revision cheat sheet" on one page. Divide the page into 6 boxes, one for each nutrient (Carbs, Protein, Fats, Vitamins, Minerals, Water). In each box, write only the *most important* information:
 1. Chemical Elements (e.g., CHO, CHON).
 2. Building Blocks (if any).
 3. Main Function(s).
 4. One Key Deficiency Disease.
 5. Two Local Food Sources.
- Use this sheet to study for your exam.



SS1 FOOD AND NUTRITION LESSON NOTES

SECOND TERM:

WEEK 1: SCIENTIFIC STUDY OF FOODS

Topic: Measurements, Units, and Accuracy

SCIENTIFIC STUDY OF FOODS: HANDY KITCHEN MEASURES



TEASPOON
(tsp) = 5ml



TABLESPOON
Tbsp = 15ml



CUP
(cup) = 250ml



**FOR ACCURACY, USE A
DIGITAL KITCHEN SCALE**

Measurements and Units

- **Introduction:** Food and Nutrition is a science. Like all sciences, it relies on accurate measurement for recipes (cookery) and experiments (food science).
- **Why We Measure:**
 - **Accuracy & Consistency:** To ensure a recipe or experiment gives the same result every time.
 - **Cost Control:** To avoid wasting ingredients.
 - **Nutritional Analysis:** To know the exact amount of nutrients in a portion.
 - **Standardization:** To create a standard recipe that anyone can follow.
- **Standard Units of Measurement:**
 - **Weight (Mass):** Measures the heaviness of an ingredient.
 - **Units:** Grams (g) and Kilograms (kg).
 - 1000 grams (g) = 1 kilogram (kg)
 - *Used for:* Solid ingredients like flour, sugar, meat, yam.
 - **Volume:** Measures the amount of space an ingredient occupies.
 - **Units:** Millilitres (ml), Litres (l), and Cubic Centimetres (cc).
 - 1000 millilitres (ml) = 1 litre (l)
 - 1 ml = 1 cc
 - *Used for:* Liquid ingredients like water, milk, oil.
 - **Temperature:** Measures the hotness or coldness.
 - **Units:** Degrees Celsius (°C) or Fahrenheit (°F).
 - *Used for:* Oven settings, water temperature, food safety (internal temp of meat).

Food Weighing and Nutrient Weight

- **Food Weighing:** The act of using a scale to determine the mass (weight) of a food item.
- **Nutrient Weight:** This refers to the mass of the specific nutrients *within* the food.
 - **Example:** If you weigh **100g** of yam (Food Weight), it is *not* 100g of carbohydrate. That 100g of yam *contains* approximately **28g** of carbohydrate (Nutrient Weight), plus water, protein, fibre, etc. Food composition tables are used to find these values.
- **Accuracy vs. Precision:**

- **Accuracy:** How *close* a measurement is to the *true* value. (e.g., you need 100g, and you measure 99.9g).
- **Precision:** How *repeatable* a measurement is. (e.g., you measure the same item 3 times and get 95.1g, 95.2g, 95.1g. This is *precise*, but not *accurate* if the true weight is 100g).
- **Goal:** In cookery, we want to be both accurate and precise.

Uses of Scales, Weights, and Handy Measures

- **A. Scales (for weighing):**
 - **Spring Balance Scale:** (Common kitchen scale). A bowl sits on a spring. Good for general cooking.
 - **Balance Scale (Weighing Balance):** (Used in markets). Uses pre-set metal *weights* (e.g., 50g, 100g, 1kg) to balance against the food.
 - **Digital Scale:** (Most accurate). Uses a battery and gives an electronic readout. Essential for professional baking and food science.
- **B. Handy Measures (for approximation):**
 - Used when a scale is not available. They are *less accurate* but useful for quick home cooking.
 - **Standard Measuring Spoons:**
 - 1 Tablespoon (tbsp) = 15ml
 - 1 Teaspoon (tsp) = 5ml
 - 3 Teaspoons = 1 Tablespoon
 - **Standard Measuring Cups:**
 - 1 Cup = 250ml
 - 1/2 Cup = 125ml
 - 1/4 Cup = 60ml
 - **Other Handy Measures:**
 - **Pinch:** The amount you can hold between your thumb and forefinger (e.g., a pinch of salt).
 - **Drop:** A single drop of liquid (e.g., drop of food colouring).

Evaluation

Objective Questions:

1. The standard unit for measuring the weight of flour is: (a) Litres (b) Grams (c) Metres (d) Degrees Celsius.
2. The standard unit for measuring the volume of milk is: (a) Kilograms (b) Grams (c) Millilitres (d) Degrees.
3. 1000g is equal to: (a) 10kg (b) 1kg (c) 100kg (d) 1ml.
4. The *most* accurate type of weighing scale is: (a) Spring scale (b) Digital scale (c) Balance scale (d) Measuring cup.

- **Theory Questions:**

1. What is the difference between "Food Weight" and "Nutrient Weight"?
2. List three "handy measures" and one type of weighing scale.

Assignment

- **Practical:** Find a recipe (e.g., for pancakes, chin-chin, or puff-puff) in a cookbook or online.
 - **Part 1:** Write down the list of ingredients.
 - **Part 2:** Identify which ingredients are measured by *Weight* (g/kg) and which are measured by *Volume* (ml/l, cups, spoons).

WEEK 2: EFFECTS OF HEAT ON NUTRIENTS

Topic: How Cooking Changes Nutrients

THE EFFECTS OF HEAT ON NUTRIENTS

APPLYING HEAT TO...



Introduction

- Cooking (applying heat) is essential: it makes food safe to eat, easier to digest, and often tastes better.

- However, heat is a destructive force. It changes the chemical structure of nutrients, which can be good or bad.
- **Methods of Heat Transfer (Recap from Term 1):**
 - **Conduction:** Direct contact (e.g., frying pan).
 - **Convection:** Movement of hot liquid or gas (e.g., boiling, baking).
 - **Radiation:** Heat waves (e.g., grilling, microwave).

Effects of Heat on Nutrients

- **A. Carbohydrates:**
 - **1. Gelatinization (Starch + Hot Liquid):**
 - **Process:** Starch granules are insoluble in cold water. When heated in a liquid (e.g., water, milk), they absorb the liquid, swell up, and burst, releasing starch molecules. This thickens the liquid.
 - **Examples:** Making *eba/gari*, *akamu/pap*, *semovita*, or thickening a soup with flour.
 - **2. Dextrinization (Starch + Dry Heat):**
 - **Process:** When starch is heated *without* liquid, the long starch chains break down into smaller chains called **dextrins**. This causes a brown colour and a slightly sweet, "toasty" flavour.
 - **Examples:** Toasting bread, frying yam, browning flour for a stew.
 - **3. Caramelization (Sugar + Dry Heat):**
 - **Process:** When sugar is heated to a high temperature, it melts, decomposes, and turns brown, developing a rich, complex "caramel" flavour.
 - **Examples:** Making caramel candy, browning onions, the brown crust on a cake.
- **B. Proteins:**
 - **1. Denaturation & Coagulation:**
 - **Process:** Heat causes the coiled protein chains to *unwind* (**denaturation**). As they continue to heat, these unwound chains tangle together and *solidify* (**coagulation**). This is an *irreversible* change.

- **Examples:** Frying or boiling an egg (the liquid white/yolk becomes solid), the "shrinking" of meat when cooked, the curdling of *wara* (cheese) from milk.
- **2. Maillard Reaction (Protein + Sugar + Heat):**
 - **Process:** A complex reaction between amino acids (from protein) and sugars that occurs at high temperatures, creating a deep brown colour and a rich, savoury flavour.
 - **Examples:** The brown crust on baked bread, the seared crust on *suya* or grilled meat, a "browning" stew.
- **C. Fats and Oils:**
 - **Melting:** Solid fats (like butter or meat fat) turn into liquid oils.
 - **Smoking Point:** When heated too high, fats begin to break down, producing visible smoke and a bitter, burnt taste. This is the **smoking point**.
 - **Rancidity:** Overheating or prolonged storage can cause fats to go rancid (spoil), developing a bad smell and "off" flavour.
- **D. Vitamins:**
 - **Water-Soluble Vitamins (B and C): Easily destroyed.**
 - **Leaching:** They dissolve in cooking water (e.g., water from boiling *ugwu* or yam). This is the *biggest* loss.
 - **Heat Destruction:** Vitamin C is *very* sensitive to heat and is easily destroyed by long cooking.
 - **Fat-Soluble Vitamins (A, D, E, K): More stable.**
 - They do *not* leach into water.
 - They are fairly resistant to heat, but some (like Vitamin A) can be lost during high-heat frying.
- **E. Minerals:**
 - **Stable:** Minerals (like Calcium, Iron) are *not* destroyed by heat.
 - **Leaching:** Like water-soluble vitamins, they can *leach* (dissolve) into the cooking water. This is why it's good to use the *pot-liquor* (cooking water) from vegetables or beans in stews or gravies.

Evaluation

Objective Questions:

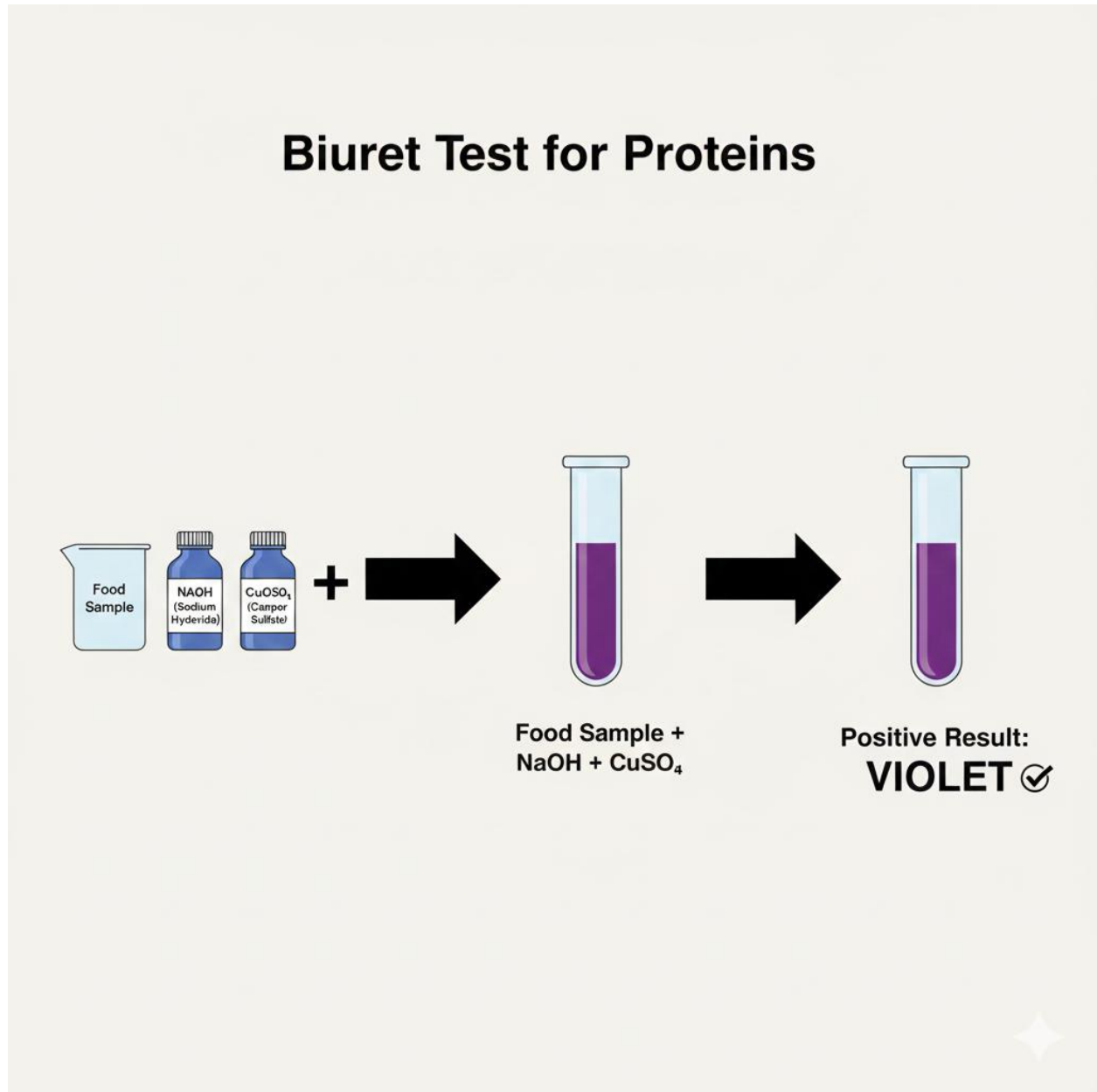
1. The thickening of *gari* when hot water is added is called: (a) Caramelization (b) Coagulation (c) Gelatinization (d) Dextrinization.
 2. The browning of bread into toast is an example of: (a) Dextrinization (b) Coagulation (c) Gelatinization (d) Melting.
 3. When an egg is fried, the protein becomes solid. This is called: (a) Caramelization (b) Coagulation (c) Gelatinization (d) Dextrinization.
 4. Which two nutrients are most easily *lost* when vegetables are boiled in water? (a) Protein and Fat (b) Vitamins B and C (c) Vitamins A and D (d) Starch and Fat.
- **Theory Questions:**
 1. Explain the difference between Gelatinization and Dextrinization. Give one food example for each.
 2. A student boils *ugwu* for 20 minutes and then throws the water away. Explain the two main ways this (bad) cooking method has caused nutrient loss.

Assignment

- Describe three different cooking methods (e.g., Boiling, Frying, Grilling). For *each* method, explain its main effect on (a) the protein and (b) the carbohydrate in a piece of yam or meat.

WEEK 3: TESTS FOR PROTEINS

Topic: Food Tests for Proteins



Introduction

- **What is a Food Test?** A chemical experiment used to detect the presence of a specific nutrient in a food sample.

- **Why test for protein?** To confirm food labels, for scientific analysis, and to understand nutrition.
- **Preparing the Sample:** Most food tests require the food to be in *liquid form*. This means solid food (like yam or beans) must first be *crushed, ground, or mashed* and then *mixed with distilled water* and *filtered* to create a liquid "food suspension."

Specific Tests for Proteins

- **A. Biuret Test (Standard Test)**
 - **Principle:** This test detects the presence of *peptide bonds* (the bonds linking amino acids). Since all proteins are made of these, it's a very reliable test.
 - **Procedure:**
 1. Place the liquid food sample in a test tube.
 2. Add an equal amount of Sodium Hydroxide (NaOH) solution (also called Caustic Soda). Shake well.
 3. *Carefully* add Copper (II) Sulphate solution, drop by drop, and shake.
 - **Positive Result:** The blue colour of the Copper Sulphate changes to **Violet or Purple**.
 - **Negative Result:** The solution remains blue.
- **B. Coagulation Test (Physical Test)**
 - **Principle:** Based on the property of *denaturation* (see Week 2). Heat or acid causes protein to solidify.
 - **Procedure (Heat):**
 1. Place a liquid food sample (e.g., liquid egg white, milk) in a test tube.
 2. Gently heat the test tube over a Bunsen burner flame.
 - **Positive Result:** The liquid protein turns into a **solid, white, or cloudy mass (a coagulum)**.
 - **Procedure (Acid):** Add a few drops of a weak acid (like lemon juice or vinegar) to milk.
 - **Positive Result:** The milk curdles (forms solid lumps).
- **C. Millon's Test**

- **Principle:** This test is *specific* for the amino acid *Tyrosine*. Most proteins contain tyrosine, so it's a good test.
- **Procedure:**
 1. Place the food sample in a test tube.
 2. Add a few drops of Millon's reagent (a solution of mercury in nitric acid).
 3. Gently heat the test tube.
- **Positive Result:** The solution or precipitate turns **Brick-Red**.
- **Note:** This test is *not* often used in schools because Millon's reagent is highly toxic (contains mercury).
- **D. Foam Test (Physical Test)**
 - **Principle:** Some proteins (like albumin in egg whites) have the ability to trap air and form a foam. This is a physical property, not a chemical test.
 - **Procedure:**
 1. Place liquid egg white in a bowl.
 2. Whisk vigorously.
 - **Positive Result:** A large, stable **foam** is produced. (This only works for a few specific proteins).

Evaluation

- **Objective Questions:**
 1. The standard chemical test for protein, which gives a purple colour, is: (a) Millon's test (b) Biuret test (c) Iodine test (d) Coagulation test.
 2. A positive Biuret test shows a colour change from blue to: (a) Red (b) Green (c) Blue-black (d) Violet/Purple.
 3. Heating liquid egg white until it becomes solid is an example of: (a) Biuret test (b) Millon's test (c) Coagulation test (d) Iodine test.
 4. Millon's test, which gives a brick-red colour, is often *avoided* because the reagent contains: (a) Acid (b) Sugar (c) Mercury (d) Iodine.
- **Theory Questions:**
 1. Why must solid food (like beans) be crushed and mixed with water before performing a Biuret test?

2. Describe the *procedure* and *positive result* for the Biuret test.

Assignment

- You are given two unlabelled test tubes. One contains a sugar-water solution, and one contains a milk solution.
- Write a plan for an experiment to determine which test tube contains the protein (milk).
- Your plan should state:
 - The *name* of the test you will use.
 - The *chemicals* you will need.
 - The *procedure* you will follow.
 - The *result* you expect to see for the milk.



WEEK 4: TESTS FOR CARBOHYDRATES, FATS, AND OILS

Topic: Food Tests for Starch, Fats, and Oils

TESTS FOR CARBOHYDRATES & FATS

STARCH TEST (IODINE)



BLUE-BLACK SPOT

Positive Result: Color change to blue-black

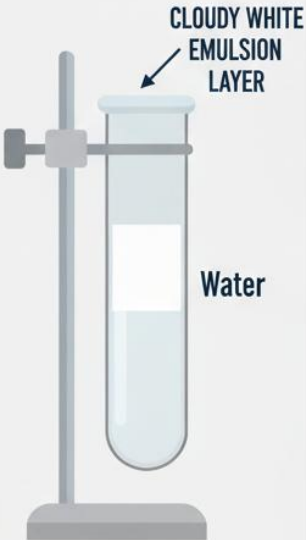
FAT TEST (PAPER)



GREASE SPOT

Positive Result: Translucent spot on paper

FAT TEST EMULSION



CLOUDY WHITE EMULSION LAYER

Water

Positive Result: Cloudy whity white emusion

Summarizes the key positive results for three different tests.

Test for Carbohydrates (Starch)

- **Iodine Test:**

- **Principle:** This is a *specific* test for **Starch**. Iodine solution reacts with the coiled starch molecule.
- **Procedure:**
 1. If the food is solid (like yam or bread), add one or two drops of Iodine solution *directly* onto a cut surface.
 2. If the food is liquid (like pap or a food suspension), place it in a test tube and add a few drops of Iodine solution.
- **Positive Result:** An intense **Blue-Black** (or dark purple) colour appears immediately.
- **Negative Result:** The iodine solution remains yellowish-brown.

Tests for Fats and Oils (Lipids)

- **A. Blotting Paper Test (Translucency or Grease Spot Test)**

- **Principle:** Fats and oils are greasy and can pass through paper, making it translucent (allowing light to pass through).
- **Procedure:**
 1. Rub or press a small piece of the food (e.g., groundnut, *suya*) onto a piece of plain white paper.
 2. Allow the paper to dry (to ensure any water evaporates).
- **Positive Result:** A permanent **translucent "grease" spot** is left on the paper.
- **Note:** This is a simple but not very sensitive test.

- **B. Sudan III Test**

- **Principle:** Sudan III is a *fat-soluble dye*. It will dissolve in fats and oils, but not in water.
- **Procedure:**
 1. Place the liquid food sample (e.g., groundnut oil and water) in a test tube.
 2. Add a few drops of Sudan III stain and shake.
- **Positive Result:** The Sudan III dye will stain the oil/fat layer, which will float on the water as a **bright red-orange layer**.

- **C. Ethanol Emulsion Test**

- **Principle:** Fats dissolve in ethanol (alcohol), but not in water. When the ethanol-fat solution is poured into water, the fat "crashes" out of the solution, forming a cloudy suspension called an **emulsion**.
- **Procedure:**
 1. Crush the food and place it in a test tube.
 2. Add about 3ml of **Ethanol** and shake vigorously to dissolve the fat.
 3. Allow the solid to settle, then *carefully* pour the clear ethanol (which now contains the fat) into a *second* test tube containing plain water.
- **Positive Result:** A **cloudy white emulsion** (looks like milk) forms in the water.
- **D. Litmus Test (Note)**
 - **Purpose:** The Litmus test is *NOT* a test for carbohydrates.
 - **What it does:** Litmus paper tests for **pH** (acidity or alkalinity).
 - Red litmus paper turns *blue* in an **alkali**.
 - Blue litmus paper turns *red* in an **acid**.
 - **Relevance:** It can be used to test if a food is acidic (like lemon juice) or if a fat has gone rancid (rancid fats produce *fatty acids*). It *cannot* be used to identify a nutrient.

Evaluation

Objective Questions:

1. The correct test for starch is: (a) Biuret test (b) Iodine test (c) Sudan III test (d) Litmus test.
2. A positive Iodine test gives what colour? (a) Purple (b) Brick-red (c) Blue-black (d) White emulsion.
3. Which test for fat involves rubbing food on paper? (a) Emulsion test (b) Sudan III test (c) Blotting paper test (d) Biuret test.
4. A cloudy white suspension formed when a food's ethanol solution is added to water indicates: (a) Starch (b) Protein (c) Sugar (d) Fat.

5. Litmus paper is used to test a food's: (a) Protein content (b) Starch content (c) pH (acidity) (d) Fat content.

- **Theory Questions:**

1. Describe the *procedure* and *positive result* for the Iodine test for starch.
2. You are given a groundnut. Describe *two* different methods you could use to prove it contains lipids (fats and oils).

Assignment

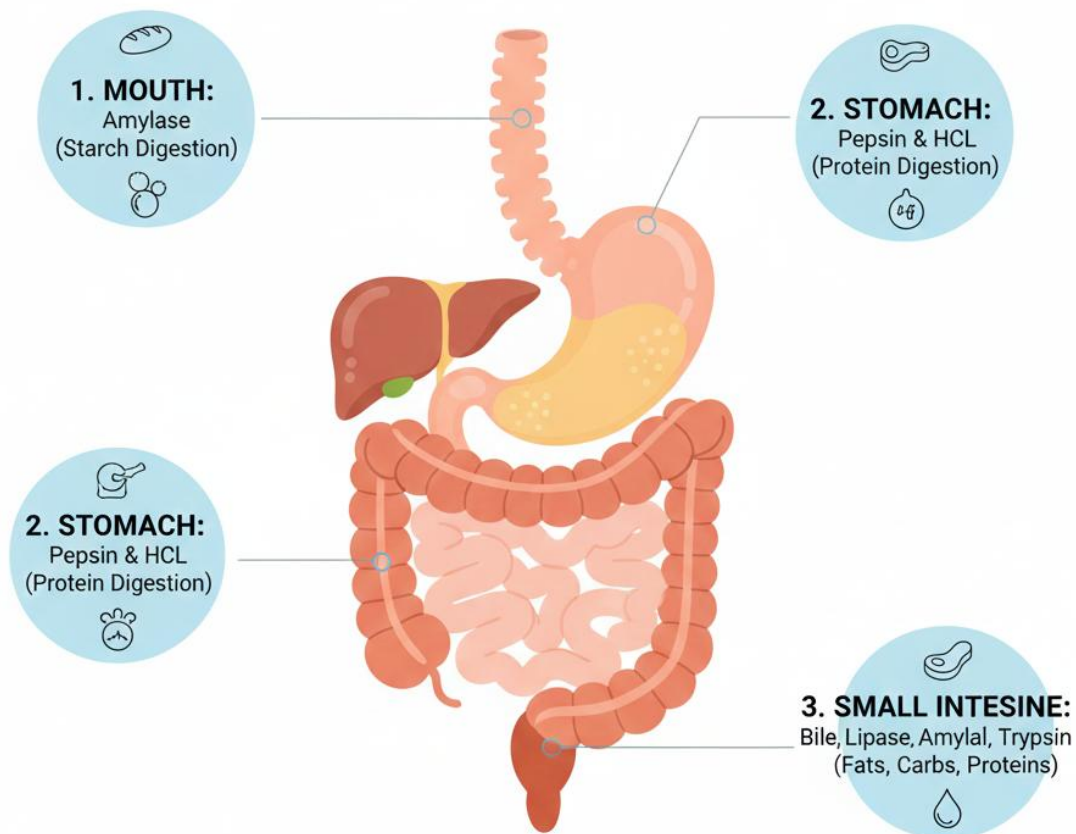
- **Practical:** Perform the **Blotting Paper Test** at home.

1. Get a piece of plain white paper.
2. Divide it into 4 sections.
3. In section 1, rub a piece of groundnut.
4. In section 2, add a drop of cooking oil.
5. In section 3, add a drop of water.
6. In section 4, rub a piece of yam or potato.
7. Let the paper dry for 30 minutes.
8. Write down your *observations* for each section. Which ones left a permanent translucent spot?

WEEK 6: THE DIGESTIVE SYSTEM

Topic: Simple Metabolic Processes and Enzymes

THE HUMAN DIGESTIVE SYSTEM: ENZYMES & ABSORPTION



Simplified Pathway Showing Main Sites of Chemical Digestion

Definitions

- **Metabolism:** The *sum of all chemical reactions* occurring in a living organism. It includes:

- **Catabolism:** Breaking down large molecules to release energy (e.g., digesting food).
- **Anabolism:** Using energy to build up large molecules (e.g., building muscle protein).
- **Digestion:** The process of **breaking down** large, complex, *insoluble* food molecules into small, simple, *soluble* molecules that can be absorbed by the body.
 - **Mechanical Digestion:** Physical breakdown (e.g., chewing, churning in the stomach).
 - **Chemical Digestion:** Breakdown by chemicals (acids and enzymes).
- **Absorption:** The process where the small, digested food molecules pass from the small intestine into the **bloodstream** or **lymph**.
- **Utilization (Assimilation):** The process where the body's cells *use* the absorbed nutrients (e.g., cells use glucose for energy, cells use amino acids to build protein).

Enzymes

- **Definition:** Enzymes are **biological catalysts**.
 - **Biological:** They are made *by* living cells (they are all proteins).
 - **Catalyst:** A substance that *speeds up* a chemical reaction *without* being used up in the process.
- **Role in Digestion:** Digestion would take *years* without enzymes. Enzymes speed it up to take *hours*.
- **Characteristics of Enzymes:**
 1. **They are specific:** Each enzyme acts on only *one* type of nutrient (its *substrate*).
 - *Amylase* only acts on *Starch*.
 - *Protease* (like Pepsin) only acts on *Protein*.
 - *Lipase* only acts on *Lipids* (fats).
 2. **They are sensitive to Temperature:** They work best at an optimal temperature (body temperature, ~37°C).
 - *Too cold:* They become inactive.

- *Too hot:* (e.g., high fever, boiling) They are *denatured* (lose their shape) and are *destroyed forever*.
3. **They are sensitive to pH:** They work best at an optimal pH (acidic or alkaline).
- *Pepsin* (in the stomach) works best at **low pH (very acidic)**.
 - *Amylase* (in the mouth) works best at **neutral pH**.
 - *Trypsin* (in the small intestine) works best at **high pH (alkaline)**.

Evaluation

- **Objective Questions:**

1. The sum of all chemical reactions in the body is: (a) Digestion (b) Absorption (c) Metabolism (d) Excretion.
2. The process of breaking down food is called: (a) Digestion (b) Metabolism (c) Absorption (d) Assimilation.
3. Biological catalysts that speed up digestion are: (a) Hormones (b) Enzymes (c) Acids (d) Villi.
4. The enzyme *Amylase* acts on: (a) Protein (b) Fat (c) Vitamins (d) Starch.
5. An enzyme's shape is destroyed by very high temperature. This is called: (a) Specificity (b) Denaturation (c) Activation (d) Absorption.

- **Theory Questions:**

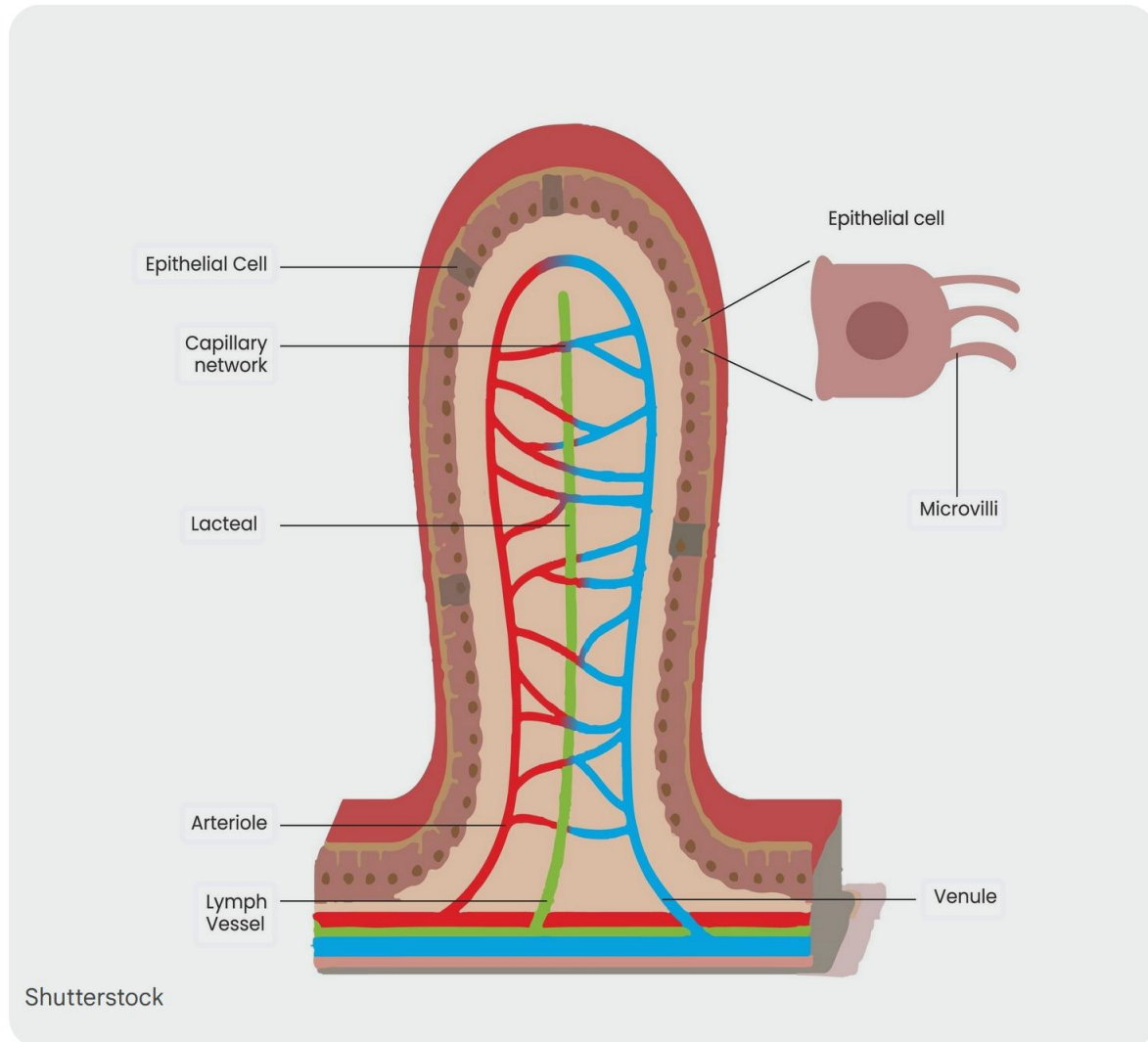
1. Differentiate between "Digestion" and "Absorption."
2. List three characteristics of enzymes.

Assignment

- Write a short paragraph explaining why a very high fever (e.g., 40°C) is dangerous. Use the word "**enzyme**" and "**denature**" in your explanation.

WEEK 8: DIGESTION AND ABSORPTION

Topic: The Journey of Food



The Digestion Process (The Alimentary Canal)

- **1. The Mouth (Oral Cavity):**
 - **Mechanical Digestion:** **Teeth** chew (masticate) food to break it into smaller pieces, increasing surface area. **Tongue** mixes food with saliva.
 - **Chemical Digestion:** **Saliva** (from salivary glands) contains the enzyme **Salivary Amylase**, which begins to digest *Starch* into *Maltose* (a simple sugar).

- The food is formed into a soft ball called a **bolus**.
- **2. The Stomach:**
 - The bolus travels down the *oesophagus* by a wave-like muscular contraction called **peristalsis**.
 - **Mechanical Digestion:** The muscular walls of the stomach *churn* and mix the food.
 - **Chemical Digestion:** The stomach lining releases **Gastric Juice**, which contains:
 - **Hydrochloric Acid (HCl):** Kills bacteria and creates a *very acidic* (low pH) environment.
 - **Pepsin:** A *protease* enzyme that begins to digest *Protein* into *Peptides* (shorter chains). Pepsin *only* works in the acidic conditions created by HCl.
 - The food is turned into a semi-liquid, acidic paste called **chyme**.
- **3. The Duodenum (First part of the Small Intestine):**
 - This is where *most* of the chemical digestion happens. The chyme enters from the stomach.
 - Two juices are added here:
 - **A. Bile:**
 - Made in the **Liver**, stored in the **Gallbladder**.
 - It is *alkaline* (neutralizes the stomach acid).
 - It **emulsifies fats** (breaks large fat globules into tiny droplets). This *increases the surface area* for lipase to work. **Bile is NOT an enzyme.**
 - **B. Pancreatic Juice:**
 - Made in the **Pancreas**. Contains a "cocktail" of powerful enzymes:
 - **Pancreatic Amylase:** Continues digesting *Starch* -> *Maltose*.
 - **Trypsin:** (a protease) Continues digesting *Protein* -> *Peptides*.
 - **Lipase:** The main fat-digesting enzyme. Digests *Fats* -> *Fatty Acids & Glycerol*.
- **4. The Ileum (Main part of the Small Intestine):**
 - **Final Digestion:** The walls of the ileum release intestinal juice, which contains:
 - *Maltase* (digests Maltose -> Glucose).
 - *Sucrase* (digests Sucrose -> Glucose + Fructose).

- *Lactase* (digests Lactose -> Glucose + Galactose).
- *Peptidases* (digest Peptides -> Amino Acids).
- **Absorption:** This is the *main job* of the ileum.
 - The inner wall is **very long** (over 6 metres) and is lined with millions of tiny, finger-like projections called **VILLI**.
 -
 - These villi give the intestine a *massive surface area* for absorption.
 - **Glucose, amino acids, vitamins B/C, and minerals** are absorbed into the **blood capillaries** in the villi.
 - **Fatty acids and glycerol** and **vitamins A, D, E, K** are absorbed into the **lacteal** (lymph vessel) in the villi.
- All undigested food (mainly **fibre**) passes into the **Large Intestine**, where water is re-absorbed, and faeces are formed.

Evaluation

• Objective Questions:

1. The enzyme in saliva, Amylase, begins the digestion of: (a) Protein (b) Fat (c) Starch (d) Vitamins.
2. The enzyme Pepsin in the stomach digests: (a) Protein (b) Fat (c) Starch (d) Vitamins.
3. Bile is produced by the ____ and its function is to ____: (a) Stomach, digest protein (b) Pancreas, digest starch (c) Liver, emulsify fats (d) Ileum, absorb nutrients.
4. The final absorption of most digested nutrients occurs in the: (a) Stomach (b) Liver (c) Ileum (Small Intestine) (d) Large Intestine.
5. The finger-like projections in the small intestine that aid absorption are called: (a) Enzymes (b) Villi (c) Bolus (d) Chyme.

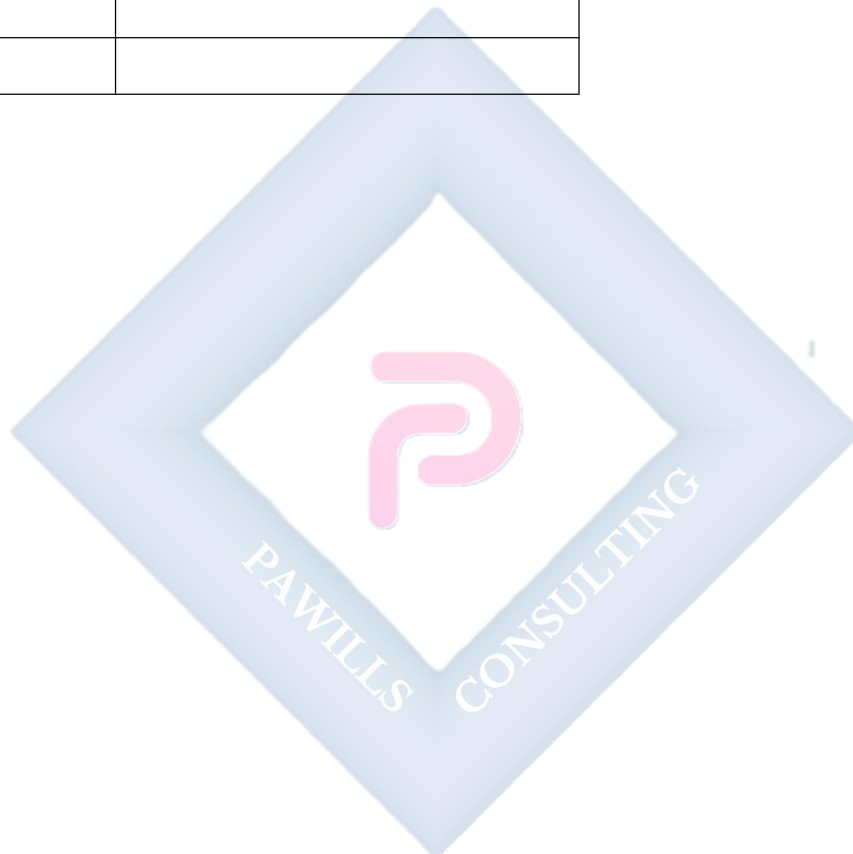
• Theory Questions:

1. What are the two main contents of Gastric Juice in the stomach, and what is the function of each?
2. Explain the function of (a) Bile and (b) Lipase in the duodenum.

Assignment

- Create a simple table with three columns: "Nutrient," "Enzyme that digests it," and "Final Product."
- Fill in the table for the three main nutrients: Starch (Carbohydrate), Protein, and Fat.

Nutrient	Enzyme(s)	Final Product(s) for Absorption
Starch		
Protein		
Fat		



WEEK 9: REPRODUCTIVE HEALTH

Topic: Nutrition and Reproductive Health

Reproductive Health: Key Nutrients in Pregnancy

Nutrient

Why it's Important



Folic Acid

Prevents neural tube birth defects, early brain & development

Makes red blood cells, prevents anemia in mother & baby



Calcium

Iron

early brain & spinal cord development

Builds strong bones & teeth, supports baby's skeletal development

Essential Vitamins & Minerals for a Healthy Pregnancy

Definitions

- **Reproductive Health:** A state of complete physical, mental, and social well-being (and not merely the absence of disease) in all matters relating to the reproductive system and its functions.
- **Reproductive Organs:**
 - **Female:** Ovaries (produce eggs, hormones), Fallopian tubes, Uterus (womb - where the baby grows), Vagina.
 - **Male:** Testes (produce sperm, hormones), Penis.

Relationship Between Nutrition and Reproductive Health

- Good nutrition is the *foundation* of reproductive health for both males and females.
- **A. Puberty:**
 - This is a period of rapid growth and development.
 - **Nutritional Needs:** Increased needs for **Protein** (for growth), **Calories** (for energy), **Calcium** (for bone growth), and **Iron** (especially for girls starting menstruation).
- **B. Menstruation:**
 - The monthly loss of blood (which is rich in **Iron**) means that females of reproductive age have a *much higher* need for iron than males.
 - **Deficiency:** Lack of iron leads to **Anaemia**, causing fatigue, weakness, and dizziness.
- **C. Fertility (Ability to conceive):**
 - **Malnutrition** (both *under* and *over*) can affect fertility.
 - **Undernutrition:** A woman who is severely underweight may stop ovulating.
 - **Overnutrition (Obesity):** Can lead to hormonal imbalances (e.g., Polycystic Ovary Syndrome - PCOS) that make conception difficult.
- **D. Pregnancy:**
 - The nutritional needs of a pregnant woman are *critical* as she is "eating for two" (growing a foetus and supporting her own body).
 - **Key Nutrients:**

- **Folic Acid (Vitamin B9):** *Crucial* in the *first few weeks* (often before a woman knows she is pregnant). It prevents **neural tube defects** (e.g., *spina bifida*) in the baby.
- **Iron:** Needs increase dramatically to make extra blood for the mother and baby.
- **Protein:** For building the baby's tissues, organs, and brain.
- **Calcium & Vit. D:** For the baby's bones and teeth.
- **Iodine:** For the baby's brain development. A severe deficiency causes **Cretinism** (permanent mental disability).
- **E. Lactation (Breastfeeding):**
 - A breastfeeding mother needs *extra calories, protein, and water* to produce nutritious breast milk.

Evaluation

- **Objective Questions:**
 1. Girls who are menstruating have a very high need for which mineral? (a) Calcium (b) Iodine (c) Iron (d) Sodium.
 2. A deficiency of which nutrient in *early* pregnancy can cause birth defects like *spina bifida*? (a) Protein (b) Iron (c) Folic Acid (d) Calcium.
 3. Severe Iodine deficiency in a mother can cause ____ in the infant. (a) Anaemia (b) Scurvy (c) Cretinism (d) Rickets.
 4. Being severely underweight or overweight can affect fertility. (a) True (b) False.
- **Theory Questions:**
 1. Why does a teenage girl's nutritional needs differ from a teenage boy's? (Mention two specific nutrients).
 2. List three nutrients that are critically important during pregnancy and explain *why* (the function) for each.

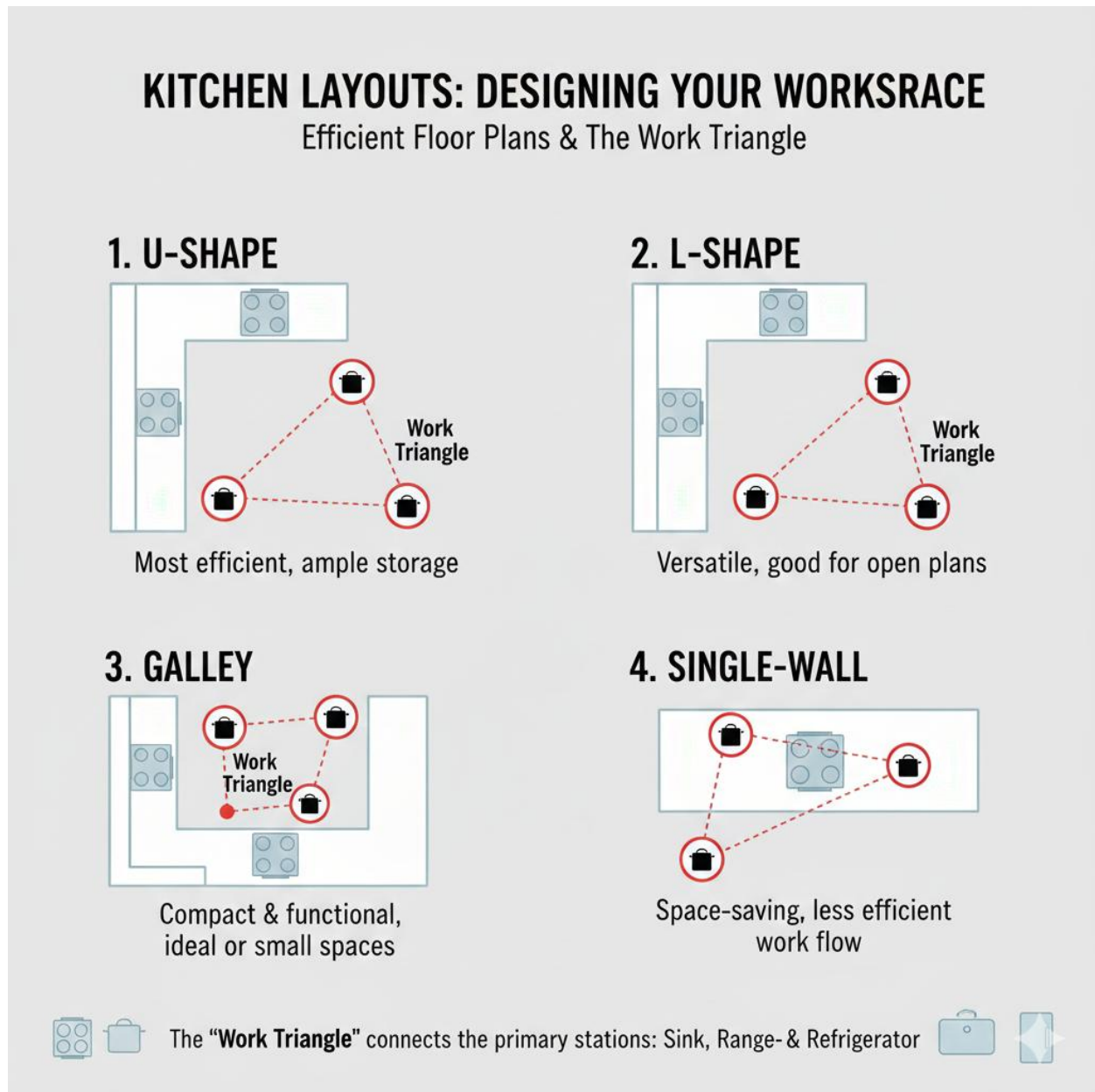
Assignment

- Write a short (2-3 paragraphs) "advice leaflet" for a newly pregnant woman. Explain to her, in simple terms, why good nutrition is important and name three specific foods she should try to eat regularly, linking them to the nutrients she needs.



WEEK 10: KITCHEN PLANS, EQUIPMENT, AND TOOLS

Topic: Kitchen Layout and Planning



Definitions

- **Kitchen:** A room or area specially designed and equipped for the storage, preparation, and cooking of food.

- **Kitchen Plan (Layout):** The arrangement of cupboards, appliances, and work surfaces in the kitchen.
- **Work Centres:** A well-planned kitchen is organized into three main "work centres" (or zones):
 1. **Storage Centre:** (Refrigerator, Freezer, Cabinets) - For storing food.
 2. **Preparation/Cleaning Centre:** (Sink, work counter, bins) - For washing, chopping, mixing.
 3. **Cooking/Serving Centre:** (Cooker, oven, microwave) - For cooking food.
- **The "Work Triangle":** A good kitchen layout tries to arrange these three centres in a small, efficient *triangle*, so the cook does not have to walk too far.

Kitchen Plans or Layout

1. **L-Shaped Plan:**
 - Appliances and cabinets are arranged along *two adjoining* walls.
 - *Pros:* Very efficient work triangle, good for open-plan homes.
2. **U-Shaped Plan:**
 - Arranged along *three* walls.
 - *Pros:* The *most efficient* layout, lots of storage and counter space, no "through traffic" to disturb the cook.
3. **Galley (Corridor) Plan:**
 - Arranged along *two parallel* walls (like a ship's galley).
 - *Pros:* Very efficient for a single cook, good for small/narrow spaces.
 - *Cons:* Can be cramped; through traffic can be a problem.
4. **Single-Wall (One-Wall) Plan:**
 - All appliances and cabinets are on *one* single wall.
 - *Pros:* Good for very small spaces (e.g., "face-me-I-face-you" apartments).
 - *Cons:* *Least* efficient, very little storage or counter space.

Factors That Affect the Size and Plan of a Kitchen

1. **Available Space:** The most important factor. You must work with the size of the room.

2. **Cost (Budget):** A larger, more complex kitchen with more cabinets is more expensive.
3. **Number of Users:** A kitchen for a large family or a restaurant needs to be bigger than one for a single person.
4. **Activities Performed:** Will the kitchen *only* be for cooking, or also for eating, laundry, and socializing?

Advantages of a Well-Planned Kitchen

1. **Saves Time and Energy:** (Efficiency) A good "work triangle" means less walking.
2. **Improves Safety:** Prevents accidents (e.g., tripping over, bumping into things).
3. **Promotes Hygiene:** Easy to clean, with no "dirt traps."
4. **More Enjoyable:** A well-organized, well-lit kitchen is a more pleasant place to work.
5. **Easy to Maintain:** Good storage means less clutter.

Evaluation

- **Objective Questions:**

1. The arrangement of appliances and cupboards in a kitchen is the: (a) Work triangle (b) Layout (c) Refrigerator (d) Cooker.
2. A good layout minimizes the distance between the sink, cooker, and: (a) Bin (b) Refrigerator (c) Door (d) Window.
3. The kitchen layout that uses three walls is the: (a) L-shape (b) U-shape (c) Galley (d) Single-wall.
4. Which layout is *least* efficient but good for small apartments? (a) L-shape (b) U-shape (c) Galley (d) Single-wall.

- **Theory Questions:**

1. What are the three main "work centres" in a kitchen?
2. List three advantages of a well-planned kitchen.

Assignment

- Draw a simple, labelled floor plan for *two* different kitchen layouts (e.g., L-Shape and U-Shape).

- On your drawings, label the three main work centres: **Cooker**, **Sink**, and **Refrigerator**.
- Draw the "work triangle" connecting them.



WEEK 11: KITCHEN PLANS, EQUIPMENT, AND UTENSILS

Topic: Selection and Care of Large Kitchen Equipment

SELECTING KITCHEN EQUIPMENT

Practical Decision Guide



COST:

Initial investment & running costs



DURABILITY:

Material quality & lifespan



SPACE OF CLEANING

Dimensions & kitchen layout fit Hygiene & maintenance



FUEL TYPE

Electric, Gas, or Other



Factors to Consider When Selecting Kitchen Equipment

1. **Cost:** What is your budget? (Buy the best quality you can afford).
2. **Durability:** Is it made of strong, long-lasting materials (like stainless steel)? Will it rust?

3. **Space:** Do you have enough space for it? (Measure the space *before* you buy).
4. **Ease of Use:** Is it simple to operate?
5. **Ease of Cleaning:** Are there "dirt traps"? Are parts removable for washing? (Hygiene).
6. **Fuel/Power Source:** Is it gas, electric, or non-electric? What is available/cheaper in your area? (e.g., Gas cooker vs. Electric cooker).
7. **Brand/Warranty:** Is it a reputable brand with a good warranty and available spare parts?

Large Kitchen Equipment – Selection and Care

- **A. Cookers (Gas, Electric, Kerosene)**

- **Selection:**

- Choose fuel type (gas is fast, electric is clean but needs reliable power).
 - Consider the number of burners and oven size needed for your family.
 - Look for safety features (e.g., flame cut-out).

- **Care:**

- Wipe up spills *immediately* to prevent stains.
 - Clean burners regularly to prevent blockage.
 - Clean the oven with an oven cleaner when dirty.
 - *Gas:* Check the gas tube for leaks regularly.
 - *Electric:* Never use water on electrical parts.

- **B. Refrigerators / Freezers**

- **Selection:**

- **Size:** Big enough for your family, but not so big it wastes energy.
 - **Energy Rating:** Look for a model with a high energy-efficiency rating (saves money on electricity bills).
 - **Features:** Do you need a separate freezer? An ice-maker?

- **Care:**

- **Do not overload:** Allow air to circulate.
 - Clean the inside regularly with water and bicarbonate of soda.
 - Clean the *outside* (coils, back) to remove dust.
 - **Defrost** the freezer section regularly if it is not "frost-free."
 - Keep the door seal (gasket) clean.

- **C. Kitchen Cabinets and Work Surfaces**

- **Selection (Work Surfaces):**

- **Laminate:** (e.g., Formica) Cheap, easy to clean, but can be damaged by heat or knives.
 - **Wood:** Looks good, but needs oiling and can stain/harbour germs.
 - **Granite/Quartz:** Very durable, heat-proof, but very expensive.

- **Care:**

- Wipe surfaces *before* and *after* every use with a clean, damp cloth.
 - **Always use a chopping board.** Never cut directly on the surface.
 - **Always use a pot stand.** Never place a hot pot directly on the surface.

Evaluation

- **Objective Questions:**

1. Which of these is *not* a factor to consider when buying a cooker? (a) Cost (b) Fuel type (c) Ease of cleaning (d) The colour of your car.
2. A refrigerator with a good energy rating will: (a) Cost more to run (b) Save money on electricity (c) Be smaller (d) Need more repairs.
3. When choosing a work surface, "Laminate" is: (a) Cheap but easily damaged by heat (b) Very expensive but durable (c) Made of wood (d) A type of cooker.

- **Theory Questions:**

1. List five general factors to consider when selecting *any* piece of kitchen equipment.
2. Describe two important "Care" rules for a refrigerator.

Assignment

- Imagine you are buying a new **Cooker**. Write two paragraphs.
 - **Paragraph 1:** Explain the *factors* you would consider before buying.
 - **Paragraph 2:** Explain the *daily care* you would give it to make it last long.

WEEK 12: SMALL OR MOVEABLE TOOLS/UTENSILS

Topic: Small Utensils and Labour-Saving Devices



Small or Moveable Tools/Utensils

- **Definition:** These are the handheld tools used for preparation, cooking, and serving.
- **Categories and Examples:**

- **Preparation:** Knives, chopping board, vegetable peeler, grater, measuring cups/spoons, mixing bowls.
- **Cooking:** Pots (with lids), pans (frying pan), wooden spoons, spatula, whisk, sieve.
- **Serving:** Ladle (soup spoon), serving spoons, plates, cutlery (fork, knife, spoon).

2. Selection, Use, and Care of Small Utensils

- **Selection:**

- **Material:**

- **Stainless Steel:** (Pots, pans, knives, bowls) *Excellent* choice. Durable, does not rust, easy to clean, does not react with food.
- **Wood:** (Spoons, chopping boards) Good (does not scratch non-stick pans), but can absorb flavours/germs. Needs thorough washing/drying.
- **Plastic/Silicone:** (Spatulas, bowls) Lightweight, cheap, but can melt, stain, or crack.
- **Non-Stick:** (Pans) Good for low-fat cooking, but the coating scratches easily.
- **Fit for Purpose:** Buy the *right tool* for the job (e.g., a small paring knife for peeling, a large cook's knife for chopping).
- **Durability:** Choose tools that are sturdy, with handles firmly attached.

- **Use and Care:**

- **Use:** Use the tool *only* for its intended purpose. (e.g., Don't use a knife as a can opener).
- **Care (General):**
 - **Wash** thoroughly after *every* use with hot, soapy water.
 - **Dry** completely before storing to prevent rust or mould.
 - **Store** properly: Knives in a knife block or rack (not loose in a drawer), utensils in a drawer or jar.
- **Care (Specific):**
 - **Knives:** Keep them sharp.
 - **Wooden tools:** Do not soak in water for long periods.

- **Non-stick pans:** Use *only* wooden or silicone spoons. *Never* use metal, as it will scratch the coating.

Labour-Saving Devices

- **Definition:** These are small appliances (usually electric) that reduce the time and effort (labour) needed for a task.
- **Examples:**
 - **Electric Blender / Liquidizer:** (e.g., for blending *egusi*, tomatoes, or making smoothies).
 - **Food Processor:** A multi-function device (chops, slices, grates, mixes).
 - **Electric Whisk / Mixer:** (e.g., for beating eggs, mixing cake batter).
 - **Pressure Cooker:** (Cooks food *very fast* using high-pressure steam, e.g., for beans or tough meat).
 - **Microwave Oven:** (Heats food *very fast* using radiation).
- **Advantages:**
 - Saves time.
 - Saves physical energy and effort.
 - Can often produce a better result (e.g., a smoother blend).
- **Disadvantages:**
 - Can be expensive to buy.
 - Reliant on electricity (a problem with NEPA/PHCN).
 - Can be difficult to clean (many parts).
 - Take up storage space.

Evaluation

- **Objective Questions:**
 1. The *best* all-purpose material for pots, pans, and mixing bowls is: (a) Wood (b) Stainless Steel (c) Plastic (d) Iron.
 2. To protect a non-stick frying pan, you should use: (a) A metal spoon (b) A wooden spatula (c) A sharp knife (d) Steel wool.

3. A device that saves time and effort, like a blender or microwave, is called a: (a) Labour-saving device (b) Large equipment (c) Handy measure (d) U-shape.
 4. A 'pressure cooker' is a labour-saving device because it: (a) Chops vegetables (b) Cooks food very fast (c) Blends soup (d) Needs no electricity.
- **Theory Questions:**
 1. List three factors to consider when *selecting* small kitchen tools (e.t., spoons, knives).
 2. What is a "labour-saving device"? List two examples and one *advantage* and one *disadvantage* of using them.

Assignment

- **Project:** Go into your kitchen at home.
 - **Part 1:** List *five* different *small utensils* you can find (e.g., grater, ladle, knife).
 - **Part 2:** List *two labour-saving devices* you can find (e.g., blender, electric kettle).
 - **Part 3:** Write one paragraph on the *general care* given to these items in your home.

SS1 FOOD AND NUTRITION LESSON NOTES

THIRD TERM:

WEEK 1: CLEANING AGENTS AND ABRASIVES

Topic: Definitions, Types, and Examples of Cleaning Agents

CLEANING AGENTS AND ABRASIVES: A FUNCTIONAL GUIDE

CLEANING AGENT	WHAT IT'S GOOD FOR
 Detergent	Grease, Oils, General Dirt 
 Acid (e.gl. Vinegar, Citric Acid)	Limescale, Rust, Mineral Deposits 
 Alkali (e.g. Baking Soda, Caustic Soda)	Burnt-on Food, Baked-on Grease, Ovens 
 Abrasive (e.gl. Scouring Powder)	Tough Stains, Scuff Marks, Scrubbing Surfaces 

CHOOSE THE RIGHT CLEANER FOR THE JOB

Definitions

- **Cleaning Agent:** Any substance, usually mixed with water, that is used to remove dirt, grease, or stains. They work by making dirt soluble or easier to lift from a surface.
- **Abrasive:** A substance or agent that cleans or polishes a surface by *friction* (rubbing and scratching). Abrasives can be part of a cleaning agent or used on their own.

Types of Cleaning Agents and How They Work

1. **Water:** The "universal solvent." It is the most common cleaning agent and is the base for most other cleaners.
 - **Hard Water vs. Soft Water:** Hard water contains dissolved minerals (calcium, magnesium) which prevent soap from lathering (forming foam), creating "scum." Soft water (like rainwater) lathers easily.
2. **Detergents (Soaps and Syndets):**
 - **How they work:** They are **emulsifying agents**. A detergent molecule has two ends:
 - A *hydrophilic* head (water-loving).
 - A *hydrophobic* tail (water-hating/oil-loving).
 - The tail attaches to the grease/oil, and the head attaches to the water. This lifts the grease from the surface, breaks it into small droplets (an *emulsion*), and allows it to be washed away.
 - **Soaps:** (e.g., bar soap) Made from natural fats/oils and an alkali (e.g., caustic soda). They work poorly in hard water.
 - **Synthetic Detergents (Syndets):** (e.g., "Morning Fresh," "Mama Lemon," "Omo," "Ariel") Man-made chemical detergents. They are more effective than soap, especially in *hard water*.
3. **Solvents:** These are agents that dissolve other substances.
 - **Examples:** Kerosene (for removing grease from bike chains), turpentine (for cleaning paint brushes), ethanol (for removing some ink stains).
4. **Acids:** Good for removing *mineral* stains, rust, and *lime scale* (the hard, white deposit from hard water).
 - **Examples:** Vinegar (acetic acid), Lemon/Lime juice (citric acid).
 - **Caution:** Acids can corrode some metals and stone.

5. **Alkalis:** Good for cutting through heavy *grease* and *fat* by reacting with them (saponification).
 - **Examples:** Baking soda (a mild alkali), Caustic soda (Sodium Hydroxide - a *very strong* alkali, used in oven cleaners), Ammonia.
 - **Caution:** Strong alkalis are *extremely* corrosive and dangerous to skin.
6. **Bleaches:** Used to whiten (by oxidation), remove stains (like tea or *jollof*), and *disinfect* (kill germs like bacteria and viruses).
 - **Example:** Sodium Hypochlorite solution (e.g., "Jik").
 - **Safety:** *NEVER* mix bleach with ammonia or acid-based cleaners. This releases toxic chlorine gas, which can be deadly.

Commercial (Manufactured) Cleaning Agents

- **Liquid Detergents:** (e.g., "Morning Fresh") For washing dishes. Contains syndets.
- **Laundry Detergents:** (e.g., "Omo," "Ariel") For washing clothes. Often contain "builders" (to soften water) and enzymes (to break down protein stains).
- **Scouring Powders:** (e.g., "Vim") These are *abrasive* cleaners. They are a mix of detergent and a fine, gritty powder (like silica).
 - **Use:** For scrubbing tough, burnt-on food from metal pots, cleaning sinks, and scrubbing floors.
 - **Caution:** *Never* use on plastic, glass, non-stick surfaces, or shiny stainless steel, as it will *scratch* and dull the surface.
- **Oven Cleaners:** Contain strong *alkalis* (Caustic Soda) to dissolve burnt-on grease. Very corrosive and dangerous. Must be used with gloves in a ventilated area.
- **Bleach:** (e.g., "Jik") For whitening clothes, disinfecting floors and toilets.
- **Disinfectants:** (e.g., "Dettol," "Izal") For killing germs on surfaces.

Local (Traditional) Cleaning Agents

- **Wood Ash:** (An *alkali*) The ash from firewood contains potassium carbonate (potash), a strong alkali. It reacts with fat to form a simple soap, making it an excellent traditional pot cleaner.

- **Sand:** A strong *abrasive*. Used with water to scrub the outside of black, sooty pots.
- **Lemon/Lime Juice:** (An *acid*) Used to remove rust stains and brighten metals like brass or copper. The acid dissolves the metal oxides.
- **Vinegar:** (An *acid*) Can be used to clean glass (removes streaks), remove odours, and dissolve lime scale.
- **Salt:** A mild *abrasive*. Can be used with lemon juice to scrub chopping boards (it cleans and deodorizes).
- **Oguso**** (Palm Fruit Chaff):** The fibrous waste from palm fruit processing can be used as a sponge or scrubbing pad (an *abrasive*).
- **Crushed Eggshells:** A mild *abrasive*, can be used to clean inside bottles.

Evaluation

• Objective Questions:

1. A substance that cleans by rubbing and friction is called: (a) A detergent (b) An abrasive (c) A solvent (d) An acid.
2. "Vim" scouring powder is an example of: (a) An abrasive (b) A solvent (c) An acid (d) A local agent.
3. Bleach (like "Jik") is used to whiten and: (a) Emulsify grease (b) Disinfect (c) Remove rust (d) Make foam.
4. A good *local* or traditional abrasive for scrubbing sooty pots is: (a) Vinegar (b) Sand (c) Kerosene (d) Salt.
5. Detergents work well in hard water because they are: (a) Natural (b) Syndets (c) Acids (d) Local.

• Theory Questions:

1. What is the main difference between a detergent and an abrasive?
2. You have a pot with burnt-on *jollof* rice. Would you use a *scouring powder* (*Vim*) or *liquid detergent* (*Morning Fresh*)? Explain your choice.
3. Explain the *scientific* reason why wood ash is a good cleaning agent for greasy pots.
4. Why should you *never* mix bleach and an acid-based cleaner?

Assignment

- Ask your parents or grandparents about the *local* cleaning agents they used before commercial products became common.
- Write a short report listing at least two local agents, what they were used to clean, and (if possible) how they were used.
- **Research:** Why does soap not lather well in "hard water," but a detergent like "Morning Fresh" does?

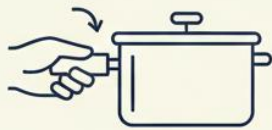


WEEK 2: SAFETY HABITS IN THE KITCHEN

Topic: Causes of Accidents; Prevention of Falls, Burns, and Scalds

SAFETY HABITS: KITCHEN FALLS, BURNS & SCALDS

✓ DO's



Turn Pot Handles Inwards
(Prevent Falls)



Use Dry Cloths/Oven Mitts
(Prevent Scalds)

DON'ts ✗



Put Wet Food in Hot Oil
(Explosion Risk)



**NEVER Use Water on a
a Grease Fire**
(Use Lid or Exthignueer)

Causes of Accidents in the Kitchen

- An **accident** is an unplanned, unfortunate event that can cause injury or damage. The kitchen is one of the most dangerous rooms in the house.

- **Common Causes:**

- **Human Factors:**

- **Haste/Rushing:** Leads to carelessness, spills, and cuts.
 - **Carelessness/Distracted:** Not paying attention, (e.g., leaving a pot to boil over, being on the phone, watching TV).
 - **Ignorance:** Not knowing how to use equipment (e.g., pressure cooker) or handle food safely.
 - **Fatigue:** Being too tired to work safely, leading to poor judgment.
 - **Poor Judgment:** (e.g., using a knife to open a can, wearing loose, flammable clothing).

- **Environmental Factors:**

- **Poor Kitchen Plan:** (See Term 2) A cramped or cluttered kitchen with an inefficient "work triangle."
 - **Poor Lighting:** Not being able to see clearly.
 - **Untidiness:** Clutter on floors and counters.
 - **Faulty Equipment:** (e.g., frayed electrical cords, leaking gas tubes).

Measures to Prevent Falls

- **Wipe up all spills (water, oil, food) *immediately*.** A wet floor is the #1 cause of falls.
- Keep floors dry and free from clutter (e.g., no bags, toys, or cartons).
- Use a sturdy, non-slip mat near the sink.
- Wear sensible, low-heeled shoes (not slippers, high heels, or just socks) in the kitchen.
- Use a sturdy step-stool to reach high shelves. *Never* use a rickety chair, stool, or climb on the counter.

Measures to Prevent Burns and Scalds

- **Distinction:**

- **Burn:** An injury from *dry heat* (e.g., flame, hot oven rack, hot metal pot).
 - **Scald:** An injury from *wet heat* (e.g., hot steam, hot oil, hot water).

- **Why Steam is Worse:** Steam is often *more dangerous* than boiling water. It carries a lot of "latent heat," which it releases instantly on contact, causing a deep, severe scald.
- **Prevention Measures:**
 - **Pot Handles:** Turn pot handles *inward* on the cooker, away from the edge where they can be knocked or grabbed by a child.
 - **Hot Pots:** Use *dry* oven mitts, cloths, or pot holders to handle hot items. A *wet* cloth will conduct heat instantly and cause a scald.
 - **Steam:** Open pot lids *away* from your face, using the lid as a shield to let the steam escape safely.
 - **Frying:**
 - *Never* overfill a pan with oil.
 - *Dry* food (e.g., yam, plantain) before putting it in hot oil. Wet food (water) will turn to steam instantly, causing the hot oil to *explode* and spit violently.
 - *Never* leave hot oil unattended.
 - **If oil catches fire:**
 1. Turn off the heat (if safe).
 2. *NEVER* use water. Water will sink, flash-boil, and cause a grease-fire explosion.
 3. Smother the flame by covering the pan with a metal lid or a damp (not soaking) cloth.
 4. Baking soda can also be used to smother a small grease fire.
 - **Children:** Keep children out of the kitchen or in a safe, supervised area, away from the cooker.
 - **Oven:** Stand to the side when opening a hot oven door to avoid a blast of hot air.
- **Electrical Safety:**
 - *Never* handle electrical appliances (e.g., blender, kettle) with *wet hands*.
 - Keep electrical cords *away* from water (the sink) and heat (the cooker).
 - Do not overload sockets with too many plugs.
 - Repair or replace any *frayed* or damaged electrical cords.

Evaluation

- **Objective Questions:**

1. The most common cause of falls in the kitchen is: (a) Sharp knives (b) Wet or oily floors (c) Poor lighting (d) Hot oil.
2. An injury from hot steam or water is called a: (a) Burn (b) Scald (c) Cut (d) Fall.
3. To prevent accidents, pot handles should be: (a) Turned outward (b) Turned inward (c) Removed (d) Pointing to the side.
4. When frying, you should *never* put ____ food into hot oil. (a) Dry (b) Wet (c) Hot (d) Cold.
5. If a pot of oil catches fire, you should: (a) Pour water on it (b) Cover it with a metal lid (c) Fan the flames (d) Pour flour on it.

- **Theory Questions:**

1. List three common *causes* of accidents in the kitchen.
2. Explain why you must use *dry* oven mitts (and not wet ones) to handle hot pots.
3. Why is steam often more dangerous than boiling water?
4. List two rules for electrical safety in the kitchen.

Assignment

- Design a simple **Kitchen Safety Poster**.
- Your poster must include at least *four* safety rules, covering *different* types of accidents (e.g., one for falls, one for burns, one for electrical).
- Use simple drawings or symbols to illustrate your rules.

WEEK 3: MEASURES TO PREVENT CUTS, SUFFOCATION, AND POISONING

Topic: Safety Habits (Continued)



Measures to Prevent Cuts

- **Knives:**

- Use *sharp* knives. A blunt knife requires more pressure, which means it is more likely to *slip* and cause a cut.
- Use the *correct knife* for the job (e.g., a small paring knife for peeling, a large chef's knife for chopping, a serrated knife for bread).

- Always cut on a stable *chopping board* (place a damp cloth underneath to stop it slipping). *Never* cut in your hand or on a slippery plate.
- Cut *away* from your body.
- Use the "claw grip": Keep your fingers curled *inward*, gripping the food, so the knife cuts against your knuckles, not your fingertips.
- *Never* try to catch a falling knife. Jump back and let it fall.
- *Never* leave sharp knives soaking in a bowl of soapy water where they cannot be seen.
- Wash knives separately and dry them carefully, blade pointing away.
- **Storage:** Store knives in a *knife block*, on a *magnetic rack*, or in a separate drawer, not loose with other utensils.
- **Broken Glass:**
 - Sweep up broken glass *immediately*.
 - Use a *damp* paper towel (or even a slice of bread) to pick up the tiny, "invisible" slivers.
 - *Never* use your bare hands.
 - Wrap the broken glass securely in newspaper before putting it in the bin.

Measures to Prevent Suffocation

- **Definition:** Suffocation is being unable to breathe.
- **Gas Cookers:**
 - Ensure the kitchen is *well-ventilated* (open windows or use an extractor fan) to prevent the build-up of **Carbon Monoxide (CO)**.
 - **CO** is a silent killer: a poisonous gas with no smell, caused by incomplete combustion (burning) of gas, kerosene, or charcoal.
 - If you smell *gas* (the chemical added to it), *do not* light a match or turn on a light switch (this can cause a spark). Open all windows, turn off the gas at the cylinder, and leave the room.
- **Charcoal/Firewood Stoves:** *Never* use a charcoal "jiko" or firewood stove indoors in a poorly ventilated room. They produce high levels of CO.
- **Plastic Bags:** Keep plastic bags (like nylon shopping bags or bread bags) *away* from small children, as they can pull them over their heads.

Measures to Prevent Accidental Poisoning

- **1. Chemical Poisoning:**

- **Storage of Chemicals:**

- Store *all* cleaning chemicals (bleach, insecticide, rat poison, caustic soda) *completely separate* from food.
- Ideally, store them in a high, *locked* cabinet, away from children and confused adults.
- Store them in their *original, labelled containers*.

- **The "Coke Bottle" Rule:**

- **NEVER** store dangerous chemicals (like kerosene, bleach, or pesticides) in old food or drink bottles (e.g., Coke bottles, water bottles).
- Someone (especially a child) may mistake it for a drink. This is a leading cause of accidental poisoning.

- **Pesticides:**

- Wash all fruits and vegetables thoroughly (under running water) to remove pesticide residues.
- Do not use insecticides (e.g., "Raid") near uncovered food.
- Keep "otapiapia" (local poison) far away from the kitchen.

- **2. Bacterial Poisoning (Food Poisoning):**

- This is the *other* type of poisoning, caused by germs in food. (See Week 10).
- Prevented by good hygiene (washing hands, cooking food properly).

Evaluation

- **Objective Questions:**

1. It is safer to use a sharp knife than a blunt one because: (a) Blunt knives slip easily (b) Sharp knives are cleaner (c) Blunt knives are heavy (d) Sharp knives are new.
2. The *safest* way to store kerosene is: (a) In a Coke bottle (b) In its original, labelled container (c) Under the sink with food (d) In a cup.

3. A major cause of accidental poisoning is: (a) Storing chemicals in drink bottles (b) Using a blunt knife (c) Wet floors (d) Not washing hands.
4. If you smell gas, you should: (a) Turn on the light to see (b) Light a match to find the leak (c) Open windows and turn off the gas (d) Immediately call the police.
5. Carbon Monoxide (CO) is a dangerous gas produced by: (a) Electrical sockets (b) Incomplete burning of fuel (c) Bleach (d) Broken glass.

• **Theory Questions:**

1. List three safety rules for *using* a knife.
2. List three safety rules for *storing* chemicals in the kitchen.
3. Why should you *never* store bleach in an old water bottle?
4. How do you safely clean up broken glass?

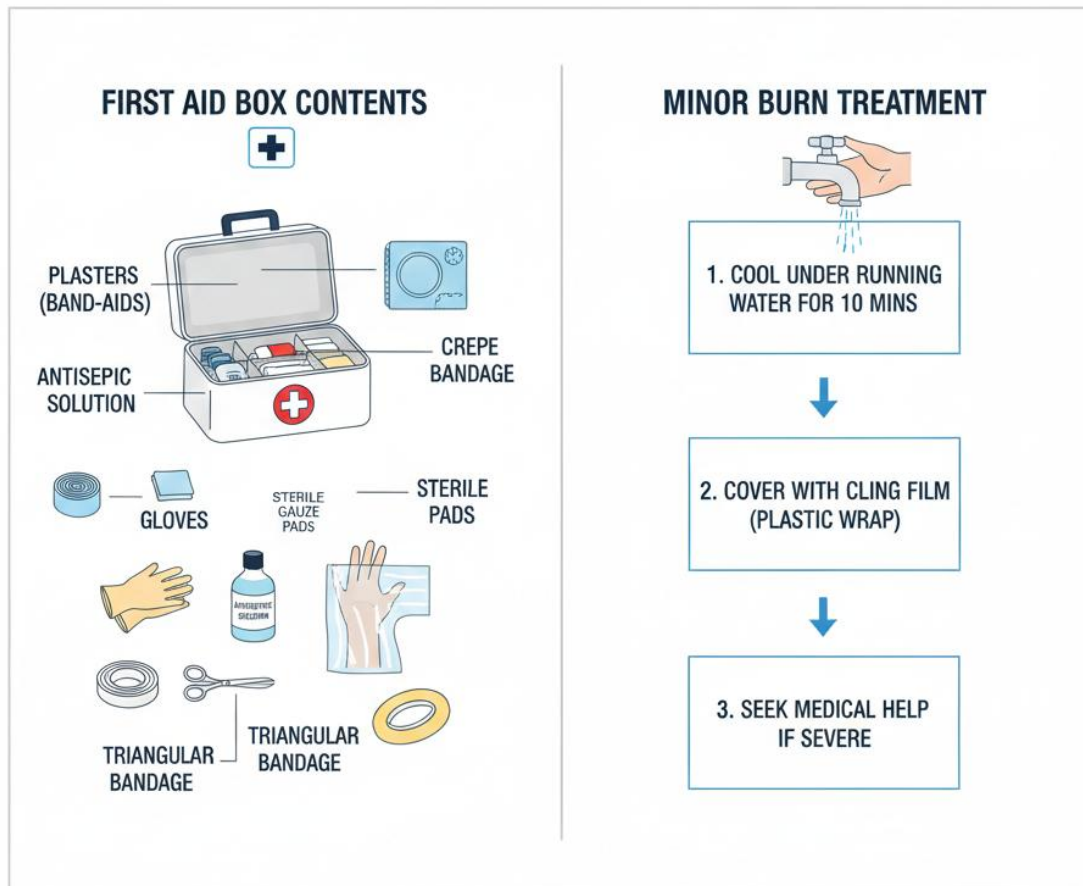
Assignment

- **Hazard Hunt:** Look around your own kitchen at home (or a kitchen you know).
- Identify *three* potential safety hazards (e.g., a wet floor, chemicals stored near food, a knife in the sink, a frayed cord, a pot handle sticking out).
- For *each* hazard, write one sentence explaining *why* it is dangerous and one sentence explaining *how* to fix it.

WEEK 4: FIRST AID BOX AND SIMPLE TREATMENTS

Topic: First Aid for Minor Kitchen Accidents

FIRST AID ESSENTIALS



Shows what to have and what to do in a common emergency.

Definition of First Aid

- **First Aid:** The *immediate* and *temporary* care given to a victim of an accident or sudden illness *before* professional medical help (a doctor or ambulance) arrives.

- **Aims of First Aid (The 3 Ps):**
 1. To **Preserve Life**.
 2. To **Prevent the condition from worsening**.
 3. To **Promote Recovery**.

Content and Use of a First Aid Box

- **Definition:** A First Aid Box is a clearly-marked, easily accessible container that holds all the essential supplies needed for giving first aid. It should be kept out of reach of children.
- **Essential Contents (and Uses):**
 - **Sterile Dressings / Gauze:** For covering larger wounds and burns.
 - **Adhesive Plasters (e.g., "Band-Aids"):** For covering small cuts and scratches.
 - **Antiseptic Wipes / Solution (e.g., Dettol, Savlon, methylated spirit):** For cleaning wounds *around* the wound to prevent infection.
 - **Bandages (Crepe and Triangular):**
 - *Crepe:* For support (sprains) or holding dressings in place.
 - *Triangular:* For making a sling (to support an injured arm).
 - **Disposable Gloves:** To protect *yourself* from blood and germs.
 - **Scissors (Blunt-ended):** For cutting bandages or clothes (safer than sharp points).
 - **Safety Pins:** For securing bandages.
 - **Calamine Lotion:** For soothing stings and rashes.
 - **Simple Painkillers (e.g., Paracetamol):** For pain relief.
 - **Tweezers:** For removing splinters.

Simple First Aid Treatment for Kitchen Accidents

- **A. Minor Cuts:**
 1. **Wash Your Hands:** Put on disposable gloves if available.
 2. **Stop Bleeding:** Apply firm, direct pressure to the wound with a clean cloth or sterile dressing for 5-10 minutes.
 3. **Clean the Wound:** When bleeding stops, clean the area with clean running water or antiseptic wipes (wipe *away* from the wound).

4. **Cover the Wound:** Apply a sterile plaster or dressing to prevent infection.
 - *For food handlers, use a blue, waterproof plaster.*
 - *See a doctor if:* The cut is deep/gaping, bleeding won't stop, there is an object (like glass) stuck in it, or it's an animal/human bite.
- **B. Minor Burns and Scalds:**
 0. **Cool the Burn:** *Immediately* place the affected area under **cool (not icy) running water** for at least **10-20 minutes**. This is the most important step to stop the burning process.
 1. **Remove Jewellery:** Gently remove any rings or watches from the affected area *before* it swells.
 2. **Cover the Burn:** Gently cover the burn with a *non-fluffy* dressing. **Cling film** (Saran wrap) is an excellent temporary covering (it's sterile and doesn't stick).
 3. **DO NOT:**
 - *Do not* burst blisters (they are a natural, sterile dressing).
 - *Do not* apply ice (can cause frostbite), butter, oil, *ugwu* leaf, toothpaste, or any other ointment. These trap heat and increase the risk of infection.
 4. **Seek Help:** For any burn larger than a N50 coin, or any burn on the face, hands, or feet, see a doctor.
- **C. Choking:** (Caused by food blocking the airway)
 0. **Encourage Coughing:** If the person is coughing, encourage them to *keep coughing*.
 1. **Back Blows:** If they cannot cough or breathe, stand behind them, lean them forward, and give **5 sharp back blows** between the shoulder blades with the heel of your hand.
 2. **Abdominal Thrusts (Heimlich Maneuver):** If that fails, stand behind them, wrap your arms around their waist, make a fist above their navel, and pull sharply *in and up* 5 times.
 3. **Repeat:** Continue 5 back blows and 5 abdominal thrusts until the object comes out or they collapse (then call for help and start CPR).

Evaluation

- **Objective Questions:**

1. Immediate, temporary care given to an accident victim is: (a) First Aid (b) Surgery (c) Nursing (d) Diagnosis.
2. Which of these is *NOT* a main aim of first aid? (a) Preserve life (b) Promote recovery (c) Give medicine (d) Prevent worsening.
3. The *first* and *most important* step for a minor burn is: (a) Apply butter (b) Cover with a cloth (c) Put it under cool running water (d) Burst the blister.
4. What should you *not* put in a First Aid box? (a) Plasters (b) Antiseptic (c) Kerosene (d) Bandages.
5. For a minor burn, you should *never* apply: (a) Cling film (b) Cool water (c) Butter or oil (d) A sterile dressing.

- **Theory Questions:**

1. List five items that should be in a kitchen First Aid Box.
2. A friend cuts their finger while chopping. What are the three steps you would take to give First Aid?
3. What are the "3 Ps" (aims) of First Aid?
4. Describe the first aid for choking.

Assignment

- **Project:** Make a "Mini First Aid Kit" for your school bag.
- Find an empty, clean container (like a small plastic box or pencil case).
- Working with your parents, try to fill it with at least *four* essential first aid items (e.g., 5-6 plasters, a small roll of bandage, antiseptic wipes, a pair of safety pins).
- Bring it to class to show.

WEEK 6: HYGIENE: PERSONAL HYGIENE

Topic: Personal Hygiene and the Food Handler

PROPER HANDWASHING STEPS



1

WET HANDS



2

APPLY SOAP



**3. SCRUB
FOR 20s**



4

RINSE



5

DRY

The single most important hygiene practice.

Definition

- **Hygiene:** The science and practice of cleanliness to maintain health and prevent the spread of disease.
- **Personal Hygiene:** The practice of cleanliness of one's own body.

Personal Hygiene Practices

- **Bathing:** Bathing or showering daily with soap and water to remove sweat, dirt, and bacteria from the skin (like *Staphylococcus aureus*).
- **Hand Washing:** The *most important* part of hygiene. Washing hands thoroughly and frequently, especially after using the toilet (to remove fecal bacteria like *E. coli*) and before touching food.
- **Oral Hygiene:** Brushing teeth at least twice a day to prevent decay and bad breath.
- **Hair:** Washing hair regularly and keeping it neat and covered when cooking.
- **Nails:** Keeping fingernails and toenails short and clean. Long nails trap a lot of dirt and bacteria.
- **Clothing:** Wearing clean clothes (including underwear) daily.

Rules for Personal Hygiene of the Food Handler

- A **food handler** is *anyone* who prepares or serves food (e.g., a chef, a *buka* owner, or you, when you cook at home).
- Their personal hygiene is *critical* because they can easily transfer pathogenic (disease-causing) germs from their body to the food, which then makes people sick.
- A person can be a "**carrier**" - they carry the germs (e.g., *Typhoid* bacteria) and can spread them, even if they don't feel sick themselves.
- **The "Dos" (What you MUST do):**
 1. **Wash Hands Thoroughly:** Use soap and running water, scrubbing for at least 20 seconds.
 - *Before* starting to cook.
 - *After* using the toilet.
 - *After* touching raw meat or fish.
 - *After* coughing, sneezing, or blowing your nose.
 - *After* touching money, a bin, or your hair/face.
 2. **Wear Clean Clothes:** Wear a clean apron or uniform.
 3. **Cover Your Hair:** Wear a hair net, chef's hat, or scarf to prevent hair from falling into food.

4. **Cover All Cuts:** Cover any cut or wound on your hands with a clean, *waterproof* plaster (often blue, so it can be seen if it falls off). Germs (like *Staph. aureus*) live in cuts.
- **The "Don'ts" (What you MUST NOT do):**
 1. **No Jewellery:** Do not wear rings (even a wedding band), watches, or bracelets. They trap dirt and germs in tiny scratches and can fall into food.
 2. **Short Nails:** Keep nails short, clean, and *unpolished* (no nail polish or false nails, as they can chip into food).
 3. **No "Tasting" with Fingers:** Do not lick your fingers or taste food from the cooking spoon and then put it back in the pot. (This transfers saliva to the food). Use a clean spoon for tasting.
 4. **No Bad Habits:** Do not smoke, spit, or chew gum in the kitchen.
 5. **Coughing/Sneezing:** If you must cough or sneeze, do it into your elbow, *not* over the food, and then *wash your hands*.
 6. **Do Not Work Sick:** *Never* handle food if you are sick, especially with diarrhoea, vomiting, or a fever. You are **highly** likely to be shedding germs.

Evaluation

- **Objective Questions:**
 1. The practice of cleanliness to maintain health is: (a) First Aid (b) Hygiene (c) Safety (d) Nutrition.
 2. The *most important* personal hygiene rule for a food handler is: (a) Wearing a hat (b) Wearing an apron (c) Frequent hand washing (d) No jewellery.
 3. A food handler should *not* wear nail polish because: (a) It is ugly (b) It is expensive (c) It can chip into the food (d) It smells bad.
 4. If you have a cut on your finger, you should: (a) Stop cooking (b) Cover it with a waterproof plaster (c) Wash it often (d) Ignore it.
 5. A "carrier" is someone who: (a) Carries food (b) Feels sick (c) Spreads germs without feeling sick (d) Wears an apron.
- **Theory Questions:**
 1. List three times a food handler *must* wash their hands.

2. List three things a food handler *must not* do in the kitchen.
3. Why should a food handler not wear jewellery (like a ring)?

Assignment

- **Handwashing Experiment:**

- **Part 1:** Before washing your hands, press your fingertips onto a piece of dark-coloured paper.
- **Part 2:** Now, wash your hands *improperly* (e.g., just with water, for 3 seconds). Press your fingertips onto another piece of paper.
- **Part 3:** Finally, wash your hands *properly* (with soap, for 20 seconds). Press your fingertips onto a third piece of paper.
- *Observe the differences* (e.g., any oily/dirty marks). Write down what this tells you about the importance of *proper* handwashing.



WEEK 8: FOOD AND KITCHEN HYGIENE

Topic: Hygiene of the Kitchen and Food



Food Hygiene

- **Definition:** The practice of handling, preparing, and storing food in a way that prevents it from becoming contaminated (with germs, chemicals, or objects) and causing illness.
- **The 4 Cs of Food Hygiene:**
 1. **Cleaning:** Keeping yourself and your kitchen clean. (See Week 6 and below).
 2. **Cooking:** Cooking food *thoroughly* to kill harmful bacteria.
 - Food (especially chicken, pork, and minced meat) should be "piping hot" all the way through (to a core temperature of 75°C).
 - Reheated food must also be "piping hot."
 3. **Chilling:** Chilling (refrigerating) food properly.
 - Cold temperatures *slow down* the growth of bacteria.
 - **The "Danger Zone":** This is the temperature range between 5°C and 63°C. In this range, bacteria multiply *very rapidly*.

- **Rule:** Keep hot food *hot* (above 63°C) and cold food *cold* (below 5°C).
- Store all perishable foods (raw meat, milk, leftovers) in the fridge (below 5°C).
- Do not leave cooked food (e.g., *jollof* rice) out in the "Danger Zone" for more than 2 hours.

4. Cross-Contamination (Preventing):

- **Definition:** The transfer of harmful bacteria from one place to another (usually from *raw* food to *cooked* food).
- **How it happens:**
 - A knife used to cut raw chicken is then used to chop salad (a ready-to-eat food).
 - Juice from raw meat at the top of the fridge drips onto a cooked stew at the bottom.
 - A dirty dishcloth used to wipe a raw meat spill is then used to wipe the counter.
- **Prevention:**
 - Use *separate* chopping boards for raw meat and ready-to-eat foods.
 - *Always* wash your hands after touching raw meat.
 - Store raw meat *at the bottom* of the fridge, in a covered container.
 - Wash knives and utensils thoroughly after they touch raw food.

Kitchen Hygiene

- **Definition:** The practice of keeping the kitchen environment, tools, and equipment clean and free from germs and pests.
- **Rules of Kitchen Hygiene:**
 - **Work Surfaces:** Clean counters *before* and *after* every use with a clean cloth and hot, soapy water (or an antibacterial spray).
 - **Dishes:** Wash dishes in hot, soapy water. Wash *cleanest* items first (glasses) and *greasiest* items last (pots). Allow to air-dry (which is more hygienic) or use a clean, dry dish towel.
 - **Cloths and Sponges:** These are "germ hotels." They are warm, wet, and full of food, making them a perfect breeding ground for bacteria. Wash and disinfect them *daily* (e.g., by soaking in bleach solution).

- **Chopping Boards:** Scrub thoroughly after every use. Use separate boards for raw meat.
- **Bins:** Keep bins *covered* with a tight-fitting lid. Empty them *daily*.
- **Floors:** Sweep daily and mop regularly.
- **Pests:** Keep food covered and the kitchen clean to prevent pests (see Week 9).

Evaluation

• Objective Questions:

1. The transfer of germs from raw food to cooked food is: (a) Cooking (b) Chilling (c) Cross-contamination (d) Cleaning.
2. To prevent cross-contamination, raw meat should be stored: (a) At the top of the fridge (b) At the bottom of the fridge (c) On the counter (d) In the oven.
3. Cooked *jollof* rice should not be left at room temperature for more than: (a) 10 minutes (b) 2 hours (c) 12 hours (d) 2 days.
4. Cold temperatures in a fridge: (a) Kill bacteria (b) Slow down bacteria growth (c) Make bacteria grow faster (d) Have no effect on bacteria.
5. The "Danger Zone" for bacterial growth is: (a) 0°C - 5°C (b) 5°C - 63°C (c) 63°C - 100°C (d) Below 0°C.

• Theory Questions:

1. What are the "4 Cs" of food hygiene?
2. Give two *specific examples* of how cross-contamination can happen.
3. What is the "Danger Zone"? Why is it important to keep food out of it?

Assignment

- **Kitchen Inspection:** Go into your kitchen and look at your chopping board(s).
- **Part 1:** What material is it made of (wood or plastic)?
- **Part 2:** Does it have many deep cuts or scratches? (These can trap bacteria).
- **Part 3:** Does your family use the same board for raw meat and vegetables?
- **Part 4:** Write a paragraph suggesting *one way* your family could improve chopping board hygiene.

WEEK 9: WASTE AND REFUSE DISPOSAL

Topic: Waste Management and Pest Control



Waste and Refuse Disposal

- **Refuse/Waste:** Any unwanted material that is discarded. In the kitchen, this includes food scraps, packaging, and liquids.
- **Methods of Waste Disposal:**
 - **Domestic (in the home):**
 - Using **refuse bins** (dustbins) lined with **bin bags**.
 - **Municipal (in the city):**

- **Collection:** Waste is collected by government (e.g., LAWMA) or private (PSP) trucks.
- **Landfill:** The most common method. Waste is taken to a large "dump site" (like Olusosun) and buried. This is problematic as it takes up space, pollutes groundwater, and releases methane (a greenhouse gas).
- **Other Methods:**
 - **Incineration:** Burning waste at very high temperatures.
 - **Recycling:** Converting waste materials (plastic bottles, metal cans, paper) into new products. This saves resources and reduces landfill.
 - **Composting:** Allowing *organic* waste (food scraps, leaves, vegetable peels) to decompose naturally and turn into *compost*, a rich fertilizer for farms and gardens. This is the best way to handle food waste.

Use and Care of Refuse Bins

- **Use:**
 - Bins in the kitchen should have a **tight-fitting lid** to keep out pests (flies, rats) and reduce smells.
 - A **bin liner (bag)** should always be used. This makes emptying the bin clean and easy.
- **Care:**
 - Empty the kitchen bin *daily*, even if it's not full, to prevent smells and pests.
 - Wash and disinfect the bin (inside and out) at least *once a week*.

Pests and Pest Control

- **Common Pests:**
 - **Rodents:** Rats, mice.
 - **Insects:** Cockroaches, flies, weevils (in flour/beans), ants.
- **Why they are bad:** They walk through sewers and dirt and then walk on food, *contaminating* it with disease-causing germs.
 - **Flies:** Carry bacteria (e.g., *Shigella* - causes dysentery, *Cholera*) from feces to food.
 - **Cockroaches:** Carry *Salmonella* and *E. coli*. Their droppings can trigger asthma.

- **Rats/Mice:** Contaminate food with urine and droppings, spreading diseases like **Lassa Fever**.
- **Pest Control (Methods):**
 1. **Deny Access (Block them):**
 - Block all small holes and cracks in walls.
 - Use mosquito netting on windows and doors.
 2. **Deny Food and Water (Starve them):**
 - This is the *most important* method.
 - *Always* keep food in sealed, covered containers (e.g., metal, hard plastic).
 - Clean up all crumbs and spills *immediately*.
 - Keep the refuse bin *covered*.
 - Do not leave dirty dishes out overnight.
 3. **Destroy Them:**
 - Use traps (e.g., mouse traps).
 - Use insecticides (e.g., "Raid") or rodenticides (rat poison).
 - **CAUTION:** These chemicals are *poisons*. They must be used *safely* and kept *far away* from all food, preparation surfaces, and children.

Evaluation

- **Objective Questions:**
 1. The *best* way to control pests in the kitchen is to: (a) Use "Raid" every day (b) Deny them food and water (c) Leave the windows open (d) Get a cat.
 2. A kitchen refuse bin should be emptied: (a) Daily (b) Weekly (c) When it smells (d) When it is full.
 3. Which disease can be spread by rats and mice? (a) Malaria (b) Lassa Fever (c) Scurvy (d) Rickets.
 4. The process of turning food scraps into fertilizer is: (a) Recycling (b) Landfill (c) Incineration (d) Composting.
- **Theory Questions:**
 1. List three rules for the proper "Use and Care" of a kitchen bin.

2. Explain the three main *principles* of pest control (Deny Access, Deny Food, Destroy).
3. Why is composting a better solution for food waste than a landfill?

Assignment

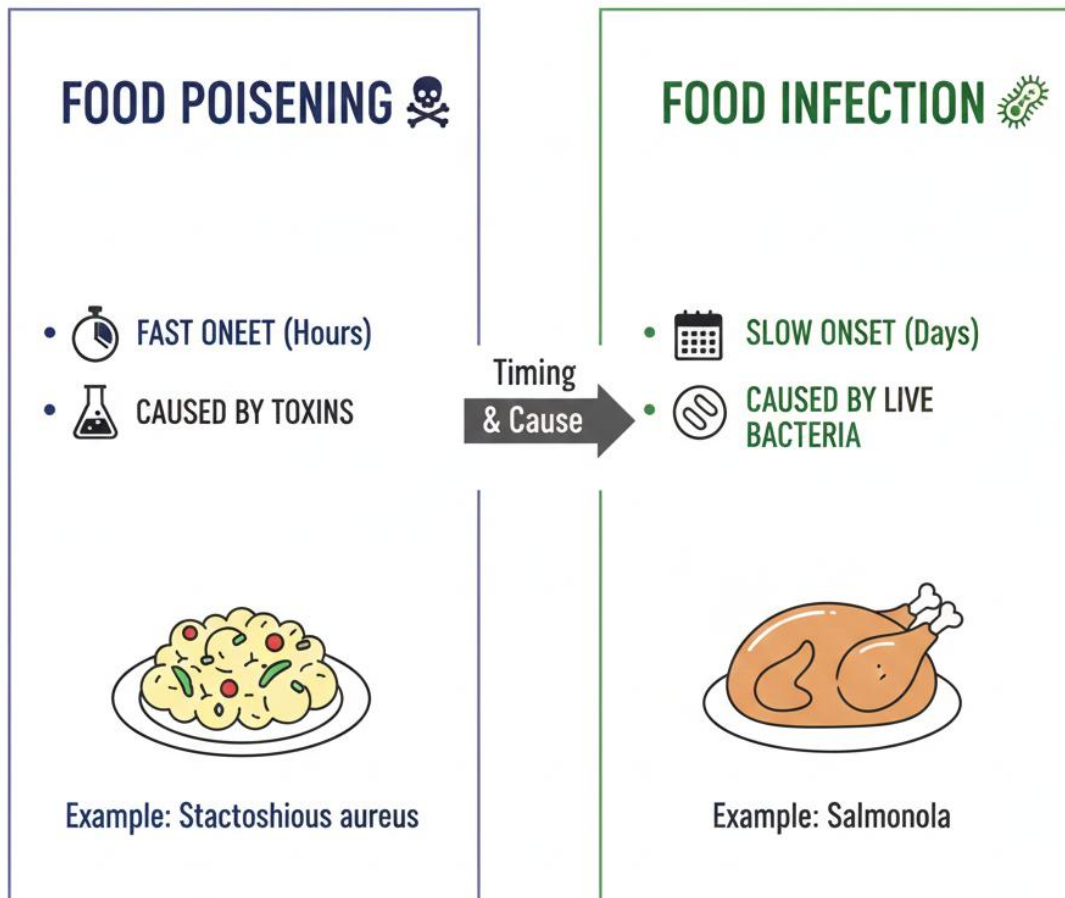
- Imagine your neighbour's kitchen has a bad problem with cockroaches.
- Write them a short, polite note explaining *three* practical steps they *must* take to get rid of the pests (without just relying on insecticide).



WEEK 10: FOOD BORNE DISEASES

Topic: Food Poisoning, Food Infection, and Prevention

FOOD-BORNE ILLNESSES: TWO TYPES



KEY DIFFERENCE: What Makes You Sick — Their Poison or Them?

Definitions

- **Food Borne Disease:** A general term for any illness caused by eating food that has been contaminated with harmful microorganisms (bacteria, viruses, parasites) or chemical toxins.
- **Food Poisoning (Intoxication):**
 - **Cause:** Caused by eating food containing *toxins* (poisons) that have been *produced* by bacteria growing in the food. The bacteria *itself* might be dead, but the poison is left. *Cooking/reheating often does NOT destroy the toxin.*
 - **Example Bacteria:**
 - ***Staphylococcus aureus*:** From unwashed hands, cuts, coughs. Toxin is heat-stable.
 - ***Bacillus cereus*:** Common in cooked rice left at room temperature. Toxin is heat-stable.
 - **Symptoms:** *Very fast* onset (1-6 hours). Main symptoms are **vomiting** and diarrhoea.
- **Food Infection:**
 - **Cause:** Caused by eating food containing the *living* harmful microorganisms, which then multiply in your gut and cause illness. *Thorough cooking KILLS the microbe and prevents the illness.*
 - **Example Bacteria:**
 - ***Salmonella*:** From raw chicken, eggs.
 - ***E. coli*:** From contaminated water, raw beef.
 - ***Listeria*:** From unpasteurized milk/cheese, can grow in the fridge.
 - ***Cholera*** (from contaminated water).
 - **Symptoms:** *Slower* onset (12-72 hours). Main symptoms are **diarrhoea**, fever, and stomach cramps.

Prevention of Food Borne Diseases

- Prevention relies on the **4 Cs** and good personal hygiene:
 - **Cleaning:** Wash hands, surfaces, and chopping boards.
 - **Cooking:** Cook food *thoroughly* (above 75°C) to kill living bacteria (*Salmonella*, *E. coli*).

- *Note:* This *will not* prevent "Food Poisoning" if the *toxin* is already in the food (e.g., *B. cereus* in rice).
- **Chilling:** Refrigerate food promptly (within 2 hours). This *stops* bacteria from multiplying and *stops* them from producing toxins. This is the best way to prevent *B. cereus* poisoning.
- **Cross-contamination (Preventing):** Keep raw and cooked foods separate to prevent the spread of *Salmonella*, etc.

Food Sanitation Laws

- **Definition:** These are government laws and regulations that all food businesses (restaurants, *bukas*, factories, street food vendors) *must* follow to ensure their food is safe for the public.
- **Enforcement:** Enforced by agencies like **NAFDAC** (National Agency for Food and Drug Administration and Control) and the Ministry of Health.
- **Typical Laws:**
 - Food handlers must be healthy and trained in hygiene.
 - The premises (kitchen) must be clean and in good repair.
 - There must be proper hand-washing facilities.
 - There must be proper waste disposal and pest control.
 - Food must be stored at the correct temperature (e.g., have a working fridge).
 - Food must come from a safe source.
 - Many modern systems use **HACCP** (Hazard Analysis Critical Control Points) - a scientific system to identify and control safety risks at every point in the food production process.

Evaluation

- **Objective Questions:**
 1. Illness from *toxins* in food, with a *fast* onset (1-6 hours), is: (a) Food Infection (b) Food Poisoning (c) Food Law (d) Food Allergy.

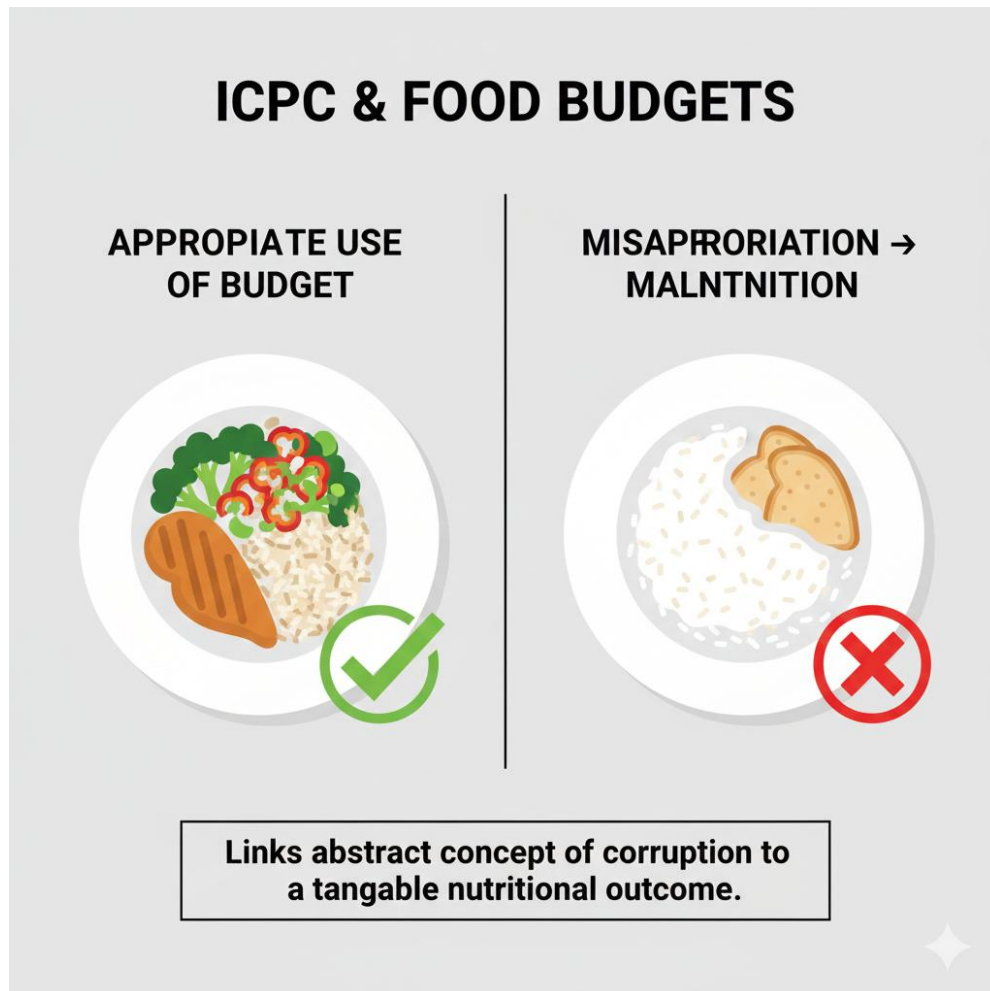
2. Illness from *living bacteria* in food, with a *slower* onset (12+ hours), is: (a) Food Infection (b) Food Poisoning (c) Food Toxin (d) Food Allergy.
 3. *Salmonella* is a type of ___ *often found in* ___. (a) Toxin, rice (b) Bacteria, raw chicken (c) Virus, water (d) Toxin, unwashed hands.
 4. The government agency that enforces food laws in Nigeria is: (a) ICPC (b) NAFDAC (c) NTA (d) LAWMA.
 5. The best way to prevent *Bacillus cereus* poisoning from cooked rice is: (a) Reheating it well (b) Chilling it quickly (c) Washing the rice (d) Adding salt.
- **Theory Questions:**
 1. What is the main difference between food poisoning and food infection? (Hint: cause and onset time).
 2. A pot of *jollof* rice is left on the counter all night after a party. Why is it dangerous to eat, even if you reheat it? (Hint: *Bacillus cereus*, toxins).
 3. What do the letters "NAFDAC" stand for?

Assignment

- **Case Study:** A student buys a chicken sandwich from a food vendor. The chicken was not cooked properly, and the vendor did not wash their hands after using the toilet.
- **Task:** Write a paragraph explaining (a) what kind of foodborne disease the student might get (Infection or Poisoning) and (b) *why* (name the bacteria and the hygiene rules that were broken).

WEEK 11: ICPC

Topic: Misappropriate Execution of Family Food Budgets



Introduction (Link to ICPC)

- **ICPC:** The "Independent Corrupt Practices and Other Related Offences Commission." This is a Nigerian government agency that fights *corruption* and *dishonesty* in public life.
- **Corruption:** Dishonest or fraudulent conduct by those in power, typically involving bribery or personal gain.
- **Connection:** Corruption is not just about government. It is about a lack of *integrity*, *honesty*, and *accountability*. These values start at home. How a family's budget is managed is a test of this character.

Misappropriate Execution of Family Food Budgets

- **Food Budget:** The amount of money a family sets aside to buy food for a period (e.g., a day, week, or month). This budget is an *act of trust*.
- **Misappropriate Execution (Misappropriation):**
 - **Definition:** This means using the food money for a purpose *other than* what it was intended for. It is a form of *dishonesty*, *theft*, and a *breach of trust* at the family level.
- **Examples:**
 1. A person is given N5,000 for the week's food. They spend only N3,000 and *lie* about the prices of food, keeping the extra N2,000 for themselves.
 2. A person in charge of cooking *deliberately* buys low-quality food (e.g., *wele* instead of meat, or rotten tomatoes) so they can "save" (steal) money from the budget.
 3. A person claims they need money for a "special soup ingredient" but uses the money to buy *airtime*.

Impact on Family Nutrition and Health

- This "domestic corruption" has severe, direct consequences:
 - **Leads to Malnutrition:** The family does not get the *quantity* or *quality* of food they need.
 - **Unbalanced Diet:** The budget holder buys only cheap *carbohydrates* (e.g., *gari*, *fufu*) and not enough *protein*, *vitamins*, or *minerals*.
 - **Child Health:** Children are the most vulnerable. This can lead to:
 - **Kwashiorkor** (protein deficiency).
 - **Anaemia** (iron deficiency).
 - Stunted growth and poor brain development.
 - **Breakdown of Trust:** It destroys trust within the family.

- **Conclusion:** The dishonesty that allows someone to steal from their family's food budget is the same mindset that allows a politician to steal from the national budget.
Accountability and **integrity** must be practised at all levels, starting at home.

Evaluation

- **Objective Questions:**

1. "ICPC" is a government agency that fights: (a) Pests (b) Food Poisoning (c) Corruption (d) Malnutrition.
2. "Misappropriation" of a food budget means: (a) Spending all the money (b) Saving money (c) Dishonestly using the money for personal gain (d) Buying food on credit.
3. A direct result of food budget misappropriation is: (a) Malnutrition (b) Better hygiene (c) More food (d) Family trust.

- **Theory Questions:**

1. Give two *specific examples* of "misappropriate execution" of a family food budget.
2. Explain how this (misappropriation) can lead to a *specific* deficiency disease in a child.

Assignment

- Imagine you are an official from the ICPC giving a talk at a PTA meeting.
- Write one paragraph explaining the "link" between stealing food money at home and national corruption. Emphasize the importance of *honesty* and *accountability* in the family.